APPLICATION NOTES

Fairchild

F8

Microprocessor

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BIT MANIPULATION

Abstract

The following application note describes the testing of various conditions (or bit pattern) in a register. It is hopeful that this will clarify the usage of bit testing in F8.

Description

There are four condition codes in F8, namely-positive, zero, overflow and carry. The user can test for any of the above four or any combination of the above four. The logical implication of these four bits are as follows:

(a) Positive: set by the most significant bit of the result being zero.

(b) Zero: set by the result being zero.

(c) Carry: set by the carry coming out of the most significant bit.

(d) Overflow: set by exclusive R the carry from the 7th bit and 8th bit. This is an indication of a change of sign in signed arithmetic.

The following are for testing bits and values of registers.

(a) Test for 1 in 6th bit of register.

```
LR    A, R
NI    B'00100000'
BNZ
```

(b) Test for 1 in 6th or 4th bit.

```
LR    A,R
NI    B'00101000'
BNZ
```

(c) Test for 1 in 6th and 4th bit.

```
LR    A, R
NI    B'00101000'
XI    B'00101000'
BZ
```

(d) Test for 6th bit in register being 0.

```
LR    A,R
NI    B'00100000'
BZ
```

(e) Test for 1 in 6th or 4th bit.

```
LR    A,R
NI    B'00101000'
XI    B'00101000'
BNZ
```

(f) Test for 0 in 6th and 4th bit.

```
LR   A,R
NI   B'00101000'
BZ
```

(g) Test to see if register is smaller than 128.

```
CI   B'10000000'
BC
```

(h) Test to see if register is greater than 254.

```
LR   A,R
CI   B'11111110'
BNC
```

For 2s complement arithmetic, the testing involves both positive and overflow bit.

(as shown in following examples).

(i) Test to see if reg R is greater than B'01110000' (+112).

```
LR   A,R
CI   B'01110000'
BM   in this case, there is no need for
     testing overflow.
```

(j) Test to see if reg R greater than B'10000000' (-128)

```
LR   A,R
CI   B'10000000'
BM   in this case, there is no need to
     test for overflow.
```

(k) Test to see if reg R is less than B'01110000' (+112).

```
        LR   A,R
        CI   B'01110000'
        BNO  Test 2
        BM   Loc
Test 2  BP   Loc
```

(l) Test to see if reg R is less than B'10000000' (-128).

```
        LR   A,R
        CI   B'10000000'
        BNO  Test 2
        BP   Loc
Test 2  BM
```

The general relation in testing 2's complement values is such that

(1) Acc $>$ value if overflow and positive are the same.

(2) Acc $\leq$ value if overflow and P are different.

(3) Overflow will not be generated unless a 2 different signed number are compared.

The following condition code table summarizes various comparisons.

Acc	Value	Result	Condition Code(OZCP)
+ve	+ve	A > B	0000
+ve	+ve	A = B	0111
+ve	+ve	A < B	0011
-ve	-ve	A > B	0000
-ve	-ve	A = B	0111
-ve	-ve	A < B	0011
+ve	-ve	A > B	0010 } (depending on operand values)
		A > B	1011
-ve	+ve	A < B	0001 } (depending on operand values)
		A < B	1000

COMPARE INSTRUCTIONS

Abstract

The following application note describes the use of the compare instructions "CM" and "CI" in performing magnitude or Algebraic comparison between two signed or unsigned numbers.

Description

F8 Microprocessor instruction set has two instructions for comparing two numbers: 'CM' and 'CI'. In 'CM' numbers compared are Memory content addressed by DC register and accumulator, while in 'CI' numbers compared are Immediate Data (second byte of instruction) and accumulator. For this discussion, accumulator content is noted by 'A' while memory content or immediate operand is noted by 'B'.

When "compare" instruction is being performed, condition codes are set of the result of following arithmetic operation: B is added with its complement of A, and 1 is added to the result.

$$\text{Result} = B + A + 1$$

Compare instructions do not affect accumulator contents.

Example	$A = 45_{16}$		0100 0111 $\quad 47_{16}$
	$B = 47_{16}$	Result =	1011 1010 $\quad$ Com 45_{16}
			1 $\quad$ Add 1
		C "1"	0000 0010

Hence the flags set would be OZCP
0011

(overflow is exclusive OR of carries from sixth and seventh stages). "0" in this example because a carry was present from the sixth and seventh stages.

As the eight bits of data in A and B can be treated as signed number (2's complement) or unsigned number (only positive numbers) it is important to know various combinations of condition codes set by different values of A and B. Following table summarizes such various combinations.

Accumulator A	Number to be compared with B	Unsigned Magnitude	Signed Magnitude	Remarks	Condition Codes OZCP
Pos.	Pos.	$A > B$	$A > B$		0000
Pos.	Pos.	$A = B$	$A = B$		0111
Pos.	Pos.	$A < B$	$A < B$		0011
Neg.*	Neg.*	$A < B$	$A < B$		0011
Neg.	Neg.	$A = B$	$A = B$		0111
Neg.	Neg.	$A > B$	$A > B$		0000
Pos.	Neg.	$A < B$	$A > B$	without OVF	0010
		$A < B$	$A > B$	with OVF	1011
Neg.	Pos.	$A > B$	$A < B$	without OVF	0001
		$A > B$	$A < B$	with OVF	1000

*Negative numbers are with Bit 7 = 1

Following ovservations can be obtained from above table.

1. If two numbers are to be compared for equality only "Z" flag is to be checked irrespective of type of numbers.

2. If unsigned magnitudes to be compared between two numbers, carry bit indicates the difference in two numbers. If carry bit is set $A \leq B$. If carry bit is not set $A > B$.

3. If signed magnitudes to be compared between two numbers (any combination) both overflow and sign bits are to be checked.

$A > B$ if both overflow and P are 1 or are zero ($0 \oplus P = 0$)

$A \leq B$ if overflow and P are not equal ($0 \oplus P = 1$)

Programs for Comparisons

1. For comparing A = B (sign or unsigned)

A is in accumulator

```
CI      B       compare with B
BZ      Equal   if zero flags set go to equal
------------    unequal
```

2. For comparing A > B (unsigned)

```
        CI      B
        BNC     Y1      if A greater than B--go to Y1
        ---------       A is either less than or equal to B

Y1
```

3. For comparing A ≥ B (unsigned)

```
        CI      B       compare with B
        BNC     Y1      If A greater than B go to Y1
        BZ      Y1      If A = B go to Y1
        ---------       Now A is less than B


Y1                      A is greater than or equal to Y1
```

4. Comparing two numbers algebraically

As per previous discussion to compare two signed numbers, exclusive or function of sign and overflow bit is required, one method of doing such function is shown below.

```
        CI      B       A is in accumulator
                        B is compared with A
        BF9     GRT     If both OVF and POS are
                        zero A > B hence branch to GRT

        BNO     Y1      To come  to this inst--either
                        overflow = 1 or Positive = 1.  If
                        overflow = Ø, positive is equal to 1
                        hence jump to Y1

        BN      GRT     Now overflow = 0, if positive =0,
                        A > B

Y1                      At this point either OVF = 1, or POS = 1
                        hence A ≤ B
```

GRT	Now OVF and Pos are both equal to zero or both equal to 1 Hence A > B.

F8 RAM EXPANSION

Abstract

A Memory Interface Chip (3852 or 3853) may be used to expand the amount of read/write memory in an F8 system up to 64K bytes. Both MI chips minimize the number of additional external parts necessary to build a working RAM interface. This paper details two system designs; one using the 3852 MI with 4K dynamic RAM, and one using the 3853 with 1K static RAM.

Static RAM Interface

The 3853 Memory Interface Chip is designed for use in interfacing various types of static RAM with an F8 system. This chip provides the following signals:

16 Address Lines - allowing up to 64K bytes of RAM expansion

CPU READ - a signal for gating data from RAM onto the F8 Data Bus lines

$\overline{\text{RAM WRITE}}$ - a memory WRITE signal

REGDR - a pin allowing external selection of the address space driven by the MI chip.

Figure 1 shows a typical system using the 3853 to connect 8K of static 2102 RAM to an F8 CPU which has 2K of F8 ROM (3851).

Connections between the 3853 and the CPU are the same as those between the F8 ROMs and CPU. Namely, 5 ROMC control lines, 8 Data Bus lines and two timing lines (Ø and WRITE) not shown on the Figure. Thus, the MI chip looks just like another ROM chip to the CPU.

Because the Data Bus lines in an F8 system are bi-directional, any chip driving this bus must be 3-state. The CMOS devices labeled 340097 on Figure 1 perform this function. Data coming from 2102 RAM is buffered through the 340097's which have an external control line for forcing them into a high output impedance state. This control line is, in turn, driven by the MI chip signal CPU READ which is ANDed with a PAGE SELECT signal derived externally. Thus, when the Data Bus Line is to be driven by RAM (during, say, an instruction fetch), the 3853

raises the CPU READ line and if the PAGE SELECT line is true, information from RAM is gated onto the Data Bus. In all other cases the CPU READ line holds the buffers in high impedance output state. Note that for RAMs which have externally controllable 3-state capability, these buffers may not be required provided the total line maximum capacitance of 100 pf. is not exceeded on the Data Bus. In this case the CPU READ signal would be connected directly to the RAM 3-state control input.

The CMOS buffers (34050) shown driving the Data Bus out to the RAM array in Figure 1 may or may not be required depending on the total line capacitance. If the capacitance on each Data Bus line can be held to under 100 pf. total, including ROM, MI, RAM inputs and interconnect, then these buffers may be eliminated.

The address space of RAM driven by the MI chip is defined externally with decoding logic. The low order ten bits of address coming from the MI are connected directly to address line inputs of the RAM chips. Drivers on the Memory Interface are designed to drive a 500 pf. load on each address line. Higher order address lines (ADDR 10 - 13) are then used in this design to determine memory pages. Each page is 1K bytes of memory. To do this, these lines are decoded by a 9315 (1 of 10 decoder) which directly generates Chip Enable signals for the RAM. Note that the first two pages of memory are used by F8 ROM (i.e. mask options have defined their ROM contents as occupying page zero and page one) and hence connot be assigned to a page of RAM.

All of the RAM chip enable signals are then effectively ORed together by the 9N30 and a PAGE SELECT signal is formed. This signal is true whenever a page of RAM is being addressed by the F8 system and is used with the CPU READ signal to gate Memory data onto the Data Bus. PAGE SELECT is also fed back into the MI chip thru an open collector buffer (shown as 9N17) tied to the REGDR pin. This pin is wire-ANDed internally (an internal pullup is also provided) with other signals and used to determine when the MI chip should be driving the Data Bus.

The system shown in Figure 1 is designed to operate at a maximum Ø clock rate of 2 MHz. To meet this, the 2102 RAM used must have a maximum access time of 650 nS.

Further RAM expansion than that shown here may be accomplished in any one of several ways. As long as the current and capacitance load driving capabilities of the various chips used is not exceeded, one simple way of further expansion is to perform more extensive decoding of the high order address lines from the MI (say perhaps with a 1 of 16 decoder, or several). RAM pages are then just stacked deeper. Another way of expanding would be to use more than one MI chip in a system. The only requirement when doing this is that the address space assigned to each MI (by the external address decoding logic) be different. In this case, each MI chip would have its own set of Data Bus buffers as well as address decoding logic and RAM array.

It should be noted here that in a system which mixes F8 ROM and Memory Interface chips, a great deal of care must be taken when using the second DC of the MI because the ROM contains only a single DC. This involves the use of the SWAP instruction which exchanges the contents of the two DC's on the MI but is a no-op to the F8 ROM since it does not have a second DC. (See the F8 Manual for further details). As a result, it may be desirable in some systems to use standard ROM chips to replace some RAM on the outside of an MI chip instead of using F8 ROMs. This gains the full advantages of the MI's dual DC but sacrafices the I/O ports, Internal timers and External Interrupt facilities of the F8 ROMs.

Dynamic RAM Interface

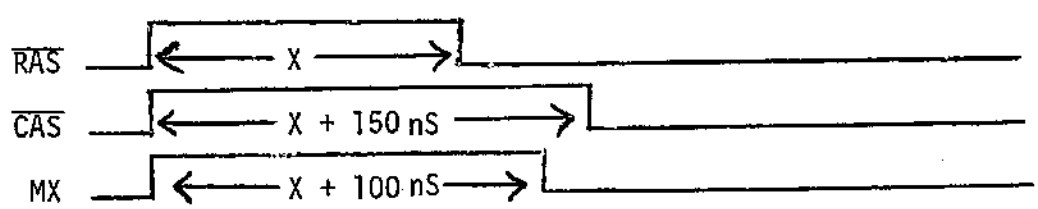
The 3852 Memory Interface Chip is designed for use in interfacing dynamic RAM with an F8 system. In addition to the signals generated by the 3853, the 3852 provides the following two signals:

CPU SLOT - used to divide a CPU instruction cycle in half. During the first half the CPU uses RAM; during the second half, RAM is refreshed.

CYCLE REQUEST - used to signal the beginning of a RAM access cycle. Each CPU cycle is divided into two sections by the CPU SLOT signal (except during execution of the ST instruction). The first section is always used for the CPU's access to RAM. The second section is then used every fourth or eighth time by the 3852 to refresh the dynamic RAM. The 3853, thus, internally multiplexes the low order Address lines between the correct RAM address (contents of the Program Counter) and an internal 6-bit refresh counter.

Figure 2 details a system design using the 3852 to connect 32K of Dynamic RAM (the 16 pin version - 4096) with an F8 CPU and 4K of F8 ROM. This design has a good deal in common with that shown in Figure 1 for static RAMs. Several other tasks must now be handled by the MI chip, however, because of the RAM refreshing requirements.

Since the 16 pin 4K RAMs are being used in this configuration it is necessary to multiplex the 12 address lines on 6 lines to be sent to the RAM chips. This may be simply accomplished using 3 9N51 (AOI) chips. The MI signal called CYCLE REQUEST is then used, thru a one-shot (9602) to generate the control gating signals for the multiplexing. CYCLE REQUEST is also used to generate the $\overline{RAS}$ and $\overline{CAS}$ strobes for the RAM chips. The timing for these signals is as follows:

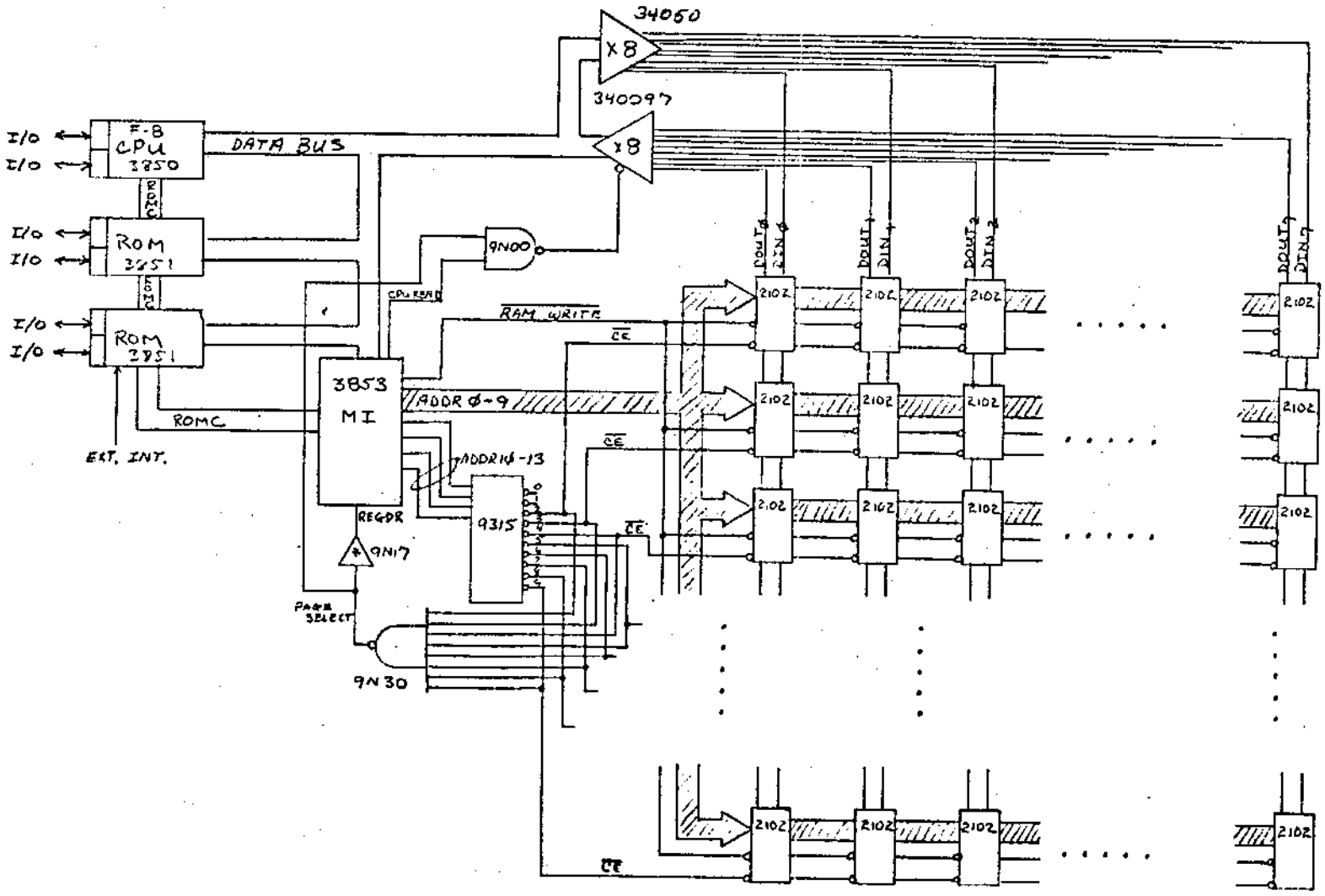


where a typical value for X is $\approx$ 300 nS and is the address set-up time on the MI chip.

During a normal CPU cycle when information is being fetched from RAM, it is necessary for the information to be held on the Data Bus lines for the entire length of the cycle. This means that in order to divide a machine cycle in half (and use the second part for refreshing RAM) it is necessary to store the information fetched during the first part of the cycle. In the design of Figure 2,

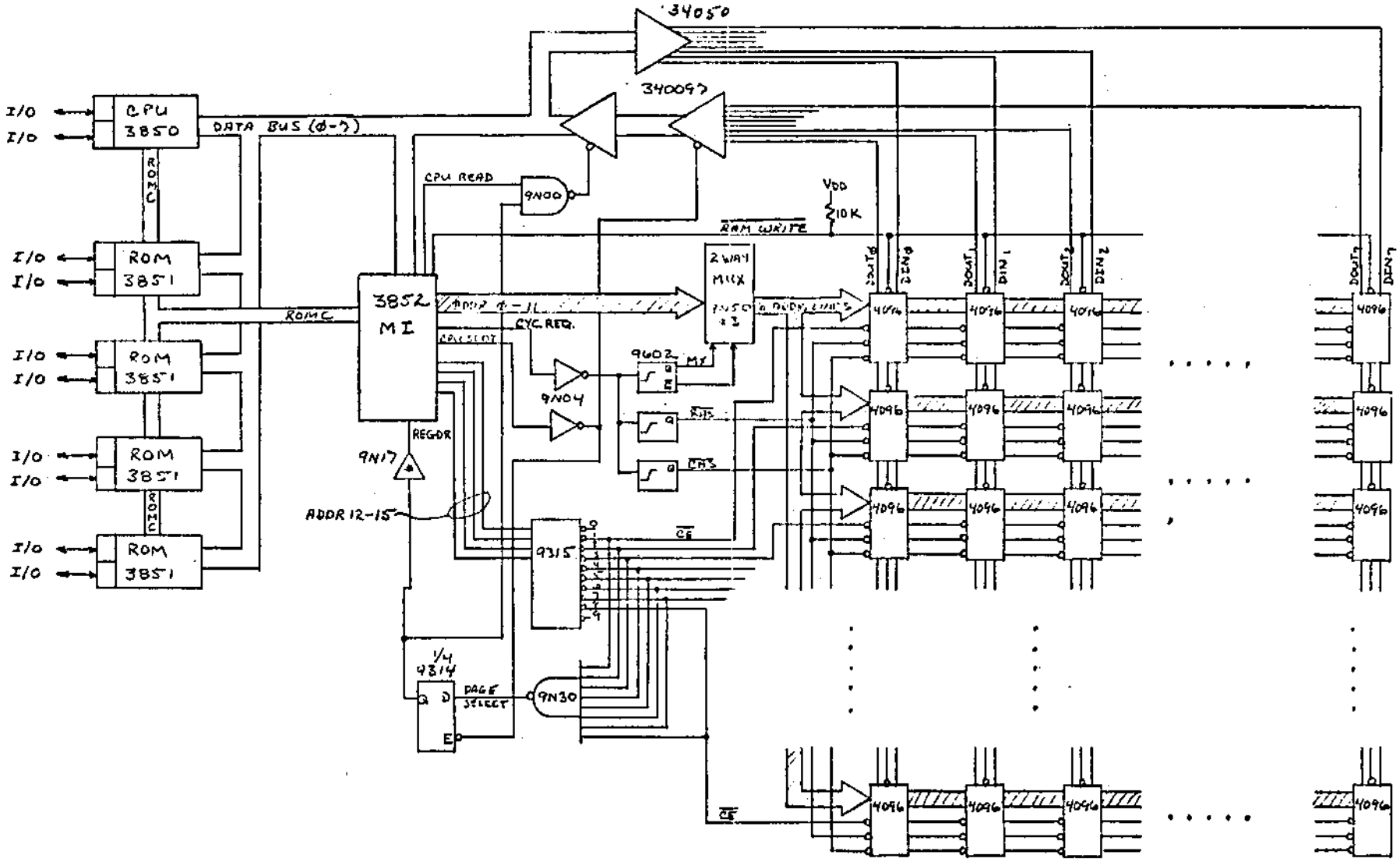
FIGURE 1

2K ROM + 8K RAM CONFIGURATION USING 2102'S



GURE 2

4K ROM + 32K RAM CONFIGURATION USING MK4096



this is accomplished for the Data Bus lines by using the CPU SLOT signal and two levels of CMOS buffering (340097's). During the first part of a cycle, CPU SLOT is true and information from RAM is passed onto the node between the two buffers. When the second part of the cycle begins (during which time RAM will be refreshed), CPU SLOT goes low temporarily latching the RAM information onto the node between the two buffers. This allows the data to remain constant on the Data Bus lines for the entire length of the cycle. An ordinary latch may also be used instead of the second level CMOS buffer described here.

Page select information must also be latched since the address lines will be changing during the second part of the cycle. One latch from a 9314 used in D-mode performs the desired function.

The system shown in Figure 2 is designed to operate with a Ø clock rate of 2 MHz. A maximum RAM access time of 350nS is required to meet this design goal.

As with the 3853 system, further expansion of RAM in a 3852 interfaced system may be accomplished in any one of several ways (further address, decoding and RAM array stacking or use of multiple MI chips). Also, the necessity of the buffers (340050) driving the Data Bus to the RAM array is contingent upon the total capacitance on those lines.

Conclusions

The 3852 and 3853 Memory Interface Chips provide an easy means of expanding the amount of Read/Write memory in an F8 system. The chips provide all timing and control signals necessary for dynamic RAM interfacing and static RAM interfacing, respectively, of most common memory chips available today. This minimizes the number of additional external circuits required and hence minimizes the overall system cost and complexity. For detailed information on the electrical and timing characteristics of the Memory Interface chips, refer to the F8 manual.

MULTIPLEXING F8 I/O PORTS

Abstract

Each I/O Port in an F8 system is 8 bits wide. Although the F8 CPU and F8 ROM each offer two I/O ports per circuit, some applications may require an expansion of I/O capability. This may be easily accomplished with standard multiplexing techniques. Three cases of I/O expansion are examined. First, consideration is given to expanding output latches of an F8 system. Next, expanding the number of inputs into an F8 is discussed. Finally, expansion of both outputs and inputs in the same system is described. All of these techniques employ MSI circuits connected to the F8 I/O port externally.

Expanding F8 Output Latches

A simple expansion of F8 I/O ports may be accomplished using two F8 I/O ports. One port will be used as the selector while the other is needed for data transfer. Using this scheme, a total of 128 output latches may be realized. The cost of this is twofold; extra circuits and additional programming.

For the purpose of analysis, this paper will expand the number of F8 output latches to eight; considerations for higher and lower expansion will also be discussed. Two types of MSI circuits are needed to accomplish this expansion. Each 8-bit I/O port will be a dual 4-bit latch. The 8-bit output latch is realized using the 9388 Dual Four-Bit Latch. A 1 of 10 decoder, the 9301, is also required.

The expansion of F8 I/O to 8 output ports is shown in Figure 1. Three leads of I/O Port 0 of the CPU are used to select one of ten ports. A fourth lead, bit 04, is used as an enable line by the latches. This bit will be used to control data entry into the latches. Port 1 of the CPU transfers data from the microprocessor to the latches. Using this method of I/O expansion, a new sequence of instructions

are necessary to output an 8-bit byte from the accumulator of the CPU to the latches. Two types of F8 instructions are used. One of these is the Load Immediate Short (LIS) instruction.

LIS a

This instruction will load the low order four bits of the accumulator with the four bit operand "a". The upper four bits of the accumulator will be set to zero. The other instruction is the Output Short instruction (OUTS) which transfers the content of the accumulator to the port specified by the four bit operand "a".

OUTS a

Each eight bit transfer to an expanded I/O latch involves five instructions. First, the data must be transferred from the accumulator to I/O port 1 of the CPU. This may be accomplished with an OUTS 1 instruction. Next, the desired latch must be selected by loading CPU port 0 with the proper code. At this time, the enable line, bit 4, must be lowered. Performing this operation allows the data to be accepted by the desired latch. This operation requires two instructions.

LIS $1X_2X_1X_0$

OUTS 0

Note: $X_2X_1X_0$ is the three bit address of the output latch.

The LIS instruction will load the accumulator with the latch address and the enable bit; the OUTS instruction transfers the content of the accumulator to I/O port 0. Finally, the enable line of the latches must be brought high, so that the contents of the latch may be maintained. This may be done by clearing port 0 as shown below:

LIS 0

OUTS 0

The sequence of instructions for an output transfer becomes:

OUTS 1	Put data on Port One
LIS $1X_2X_1X_0$	Select a latch, lower the enable line
OUTS 0	Put the select bits on port 0
LIS 0	Clear the accumulator
OUTS 0	Raise the enable line port 0

The first instruction makes the data available for all of the latches. The third instruction selects the desired latch and drops the enable line, allowing the desired latch to accept the data. The last instruction raises the enable line so that the output data is held. All five instructions are one byte each. The OUTS instruction require two cycles to execute while the LIS instruction only requires one. Thus, each output to an expanded latch is a five byte operation executing in eight cycles.

Output Macro

A macro may be developed to handle a data transfer to these external latches. Suppose that a byte of data is already in the accumulator. A suitable output macro would be:

```
MACRO
OUTPUT &SEL     Macro Name
OUTS 1          Output Data From Accumulator
LIS &SEL        Put Selection Code in Accumulator
OUTS 0          Output Selection Code to Port 0
LIS 0           Clear Accumulator
OUTS 0          Raise Latch Enable Line
MEND
```

The argument, &SEL, is the selection code for the desired latch. When used within a program, this macro will transfer the accumulator into the latch designated by SEL. The content of the accumulator is cleared after this operation, as is the

content of I/O Port 0. The data that was in the accumulator is stored in Port 1.

Example

Suppose an F8 system is configured with eight expanded output latches, as shown in Figure 1 and it is necessary to transfer the contents of scratchpad register one to latch six and scratchpad register two to latch three. The following sequence of instructions may be used.

```
LR A,1            Load R1 into the Accumulator
OUTPUT B'1110'
LR A,2            Load R2 into the Accumulator
OUTPUT B'1011'
```

Some cases may exist where the number of instructions per output may be reduced. For instance, in cases where all of the latches are to be loaded with identical data, only one output instruction to port number one is required. In another situation, where only one latch will experience data changes, that latch can be selected once, and the data in the latch may change by simply writing into I/O Port one. If all of the latches are to be cleared, the Master Reset (MR) line of these registers may be connected to a pin of port zero, the selector port. Thus, a pulse on this line will reset the latches.

Other Configurations

When a different number of latches are used in this type of output expanding technique, some system details may change. If, for instance, only seven latches or less are required, the need for a decoder is not necessary. The selector I/O port can use seven lines for latch selection and the eighth line for the enable strobe. If more than eight latches are employed, a larger decoder may be required. In addition, an extra instruction byte is required. LIS is a one byte instruction. However, if more than eight latches are used, the selection code will be at least four bits; add another bit for the enable bit and a five bit select word is require The LIS instruction only has a four bit operand. Thus, the Load Immediate instruc-

tion (LI), a two byte instruction, will be required.

Expanding F8 Inputs

An input into the F8 system may be sampled from any F8 Port. Applications exist where it may become necessary to increase the number of input sources into an F8 system. This may be done using multiplexing techniques. Once again, two F8 I/O Ports will be required. One port, CPU Port 0, will be used to select one of many input sources. The other port, CPU Port 1, will pass the data from the source to the accumulator of the CPU.

During this discussion, a system expansion to sixteen input sources will be examined. Only one type of TTL MSI circuit is required. This is a 16-input Multiplexer, the 93150.

The implementation of input expansion is rather straightforward. A diagram of the circuit is shown in Figure 2. The two CPU Ports are used. The output of port zero is connected through buffers to the input of the sixteen multiplexer decoder inputs. A buffer is required because the F8 I/O Port can only drive one standard TTL load. (Designs employing another technology, Low Power TTL or CMOS, may eliminate the need for a buffer, at the expense of other design tradeoffs). The output of the F8 I/O Ports are inverted. Thus, the external buffer will invert the output to its original state. Once the data source has been selected, an 8-bit byte is available on CPU I/O Port 1.

The sequence of F8 instructions for reading one of the sources of data from the configuration in Figure 2 is:

```
LIS   a   Select a Data Source
OUTS  0
INS   1   Read the Data
```

The four bit operand of the LIS instruction selects one of 16 data sources. Once this code has been entered into the accumulator, an output instruction transfers it to CPU Port 0. Thus, the select code will be issued to the multiplexers. The proper data is now available on Port 1 of the CPU. An input instruction, INS 1,

will transfer this data from I/O Port 1 to the accumulator. In most cases, each input to the accumulator from one of the expanded input sources requires the above three instructions. These instructions are each one byte long. The OUTS and INS instructions execute in two cycles while the LIS instruction requires only one cycle. Thus, each input from configuration of Figure 2 is three bytes long and executes in five machine cycles.

Input Macro

A macro may be written to perform an input operation into the F8. The argument, SEL, selects the source of data.

```
MACRO
INPUT &SEL
LIS &SEL      Put Select Code into Accumulator
OUTS 0        Output Select Code to Port 0
INS 1         Read Data into Accumulator
MEND
```

This macro will read data from the desired source and leave it in the accumulator. The original content of the accumulator are lost.

Example

Suppose it is necessary to read the data from sources two and nine in Figure 2 and store the data in scratchpad registers one and two respectively. The program to perform this operation would be:

```
INPUT B'0010'
LR 1,A          Store Data in Reg 1
INPUT B'1001'
LR 2,A          Store Data in Reg 2
```

First, a select code for data source two must be put in the accumulator and then transferred to I/O Port 0. Then the data may be read into the accumulator and passed on to register one of the scratchpad. This sequence of operations is repeated

for the second byte of data.

If data is continuously read from the same source, it will not be necessary to update the select code. This will save two instructions per input.

Other Configurations

As a larger number of data sources are required, the amount of multiplexing will increase proportionally. The multiplexers may be stacked to allow a total of 256 sources of data using two I/O Ports.

Expanding Inputs and Outputs Simultaneously

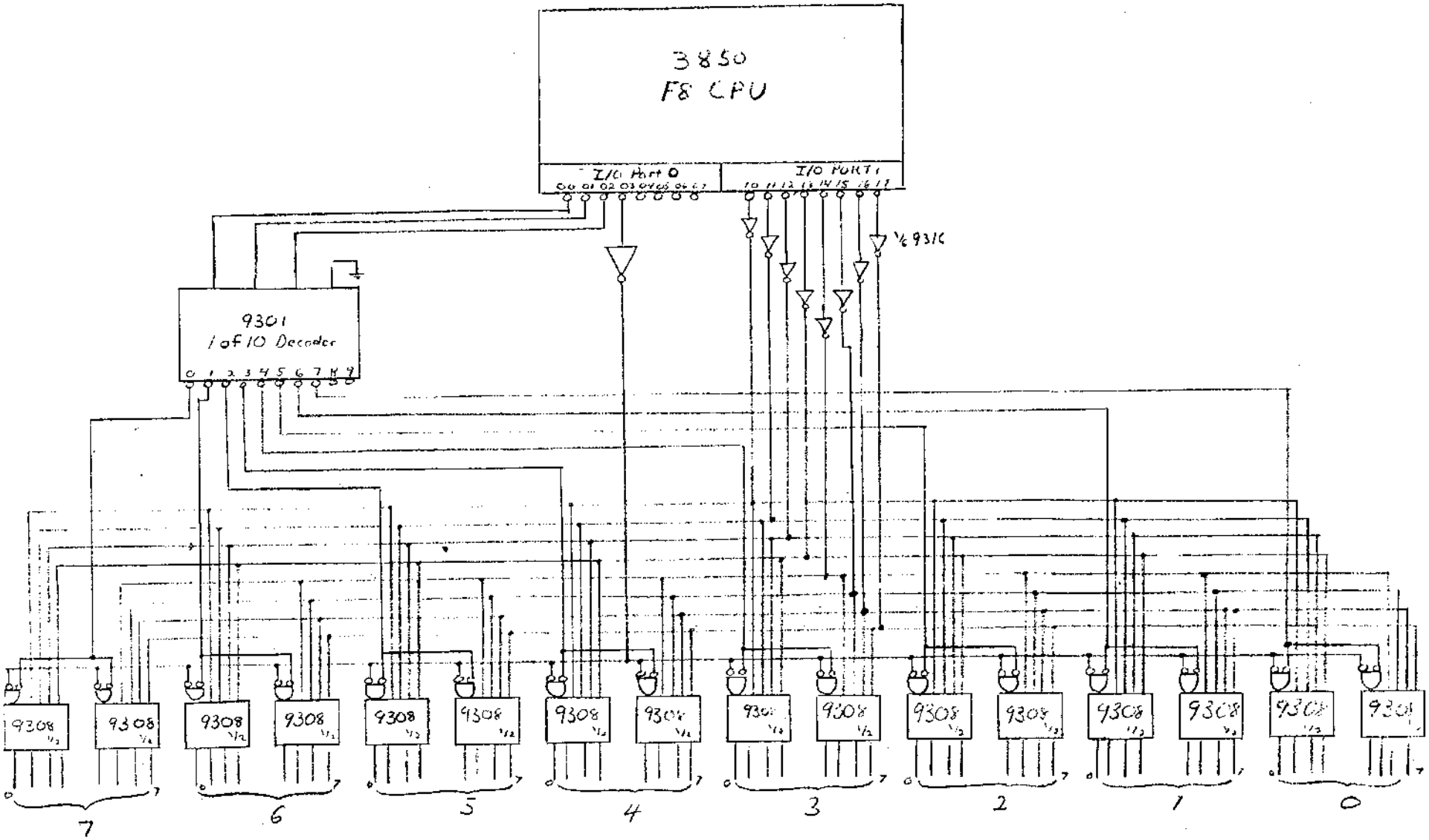
The concept of expanding the inputs and the outputs may be combined. Still using just two F8 I/O ports, a system may be configured so that the number of output latches and input sources is increased. One I/O Port will be used as a selector. In addition to choosing the source or destination of the data, the selector port must also specify the direction of data flow. The other F8 I/O Port will be used to transfer the data.

In order to combine both the input and output simultaneously, it is necessary that the input lines do not affect the logic levels of the I/O Port during the output mode. Therefore, buffering is necessary. A three-state CMOS Hex non-inverting buffer has been selected for this task.

The full expanded system is shown in Figure 3. Bit 7 of CPU I/O Port 0 controls the direction of data transfer. When it is HIGH (LOGIC ONE), data may be transferred to the output latches. When this bit is zero, the input data from the multiplexers may be sampled.

Conclusion

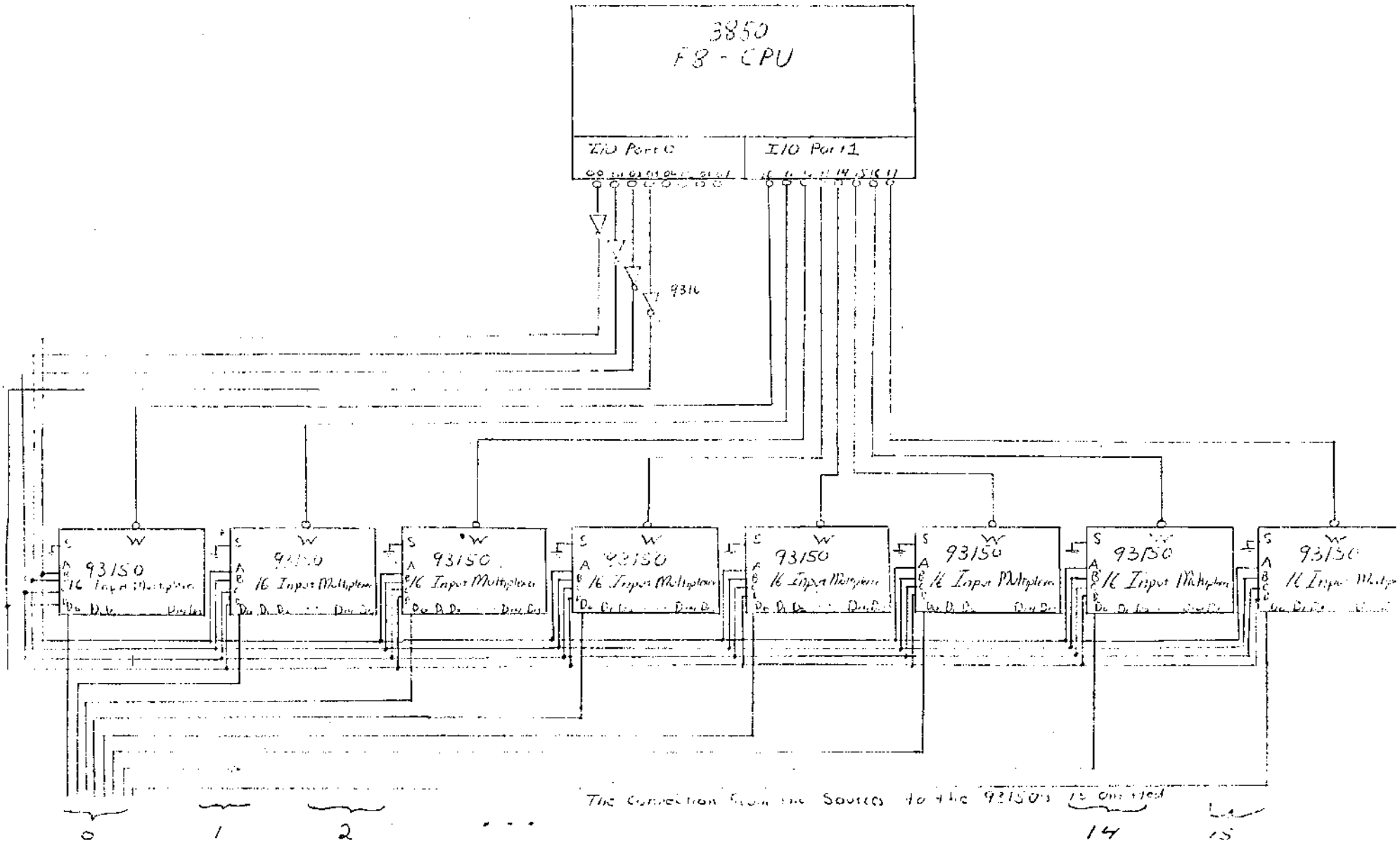
The expansion of F8 I/O ports with external MSI circuits may be accomplished using multiplexing techniques. Although this expansion requires additional programming and additional circuits, it is indeed simple. Timing considerations are kept to a minimum, in addition, the users do not have to worry about other aspects of the F8 system. The external circuits are completely independent to the microprocessor.



Part No.	Quantity
9308	8
9301	1
9316	2

Expanded Output OF F8 Latches

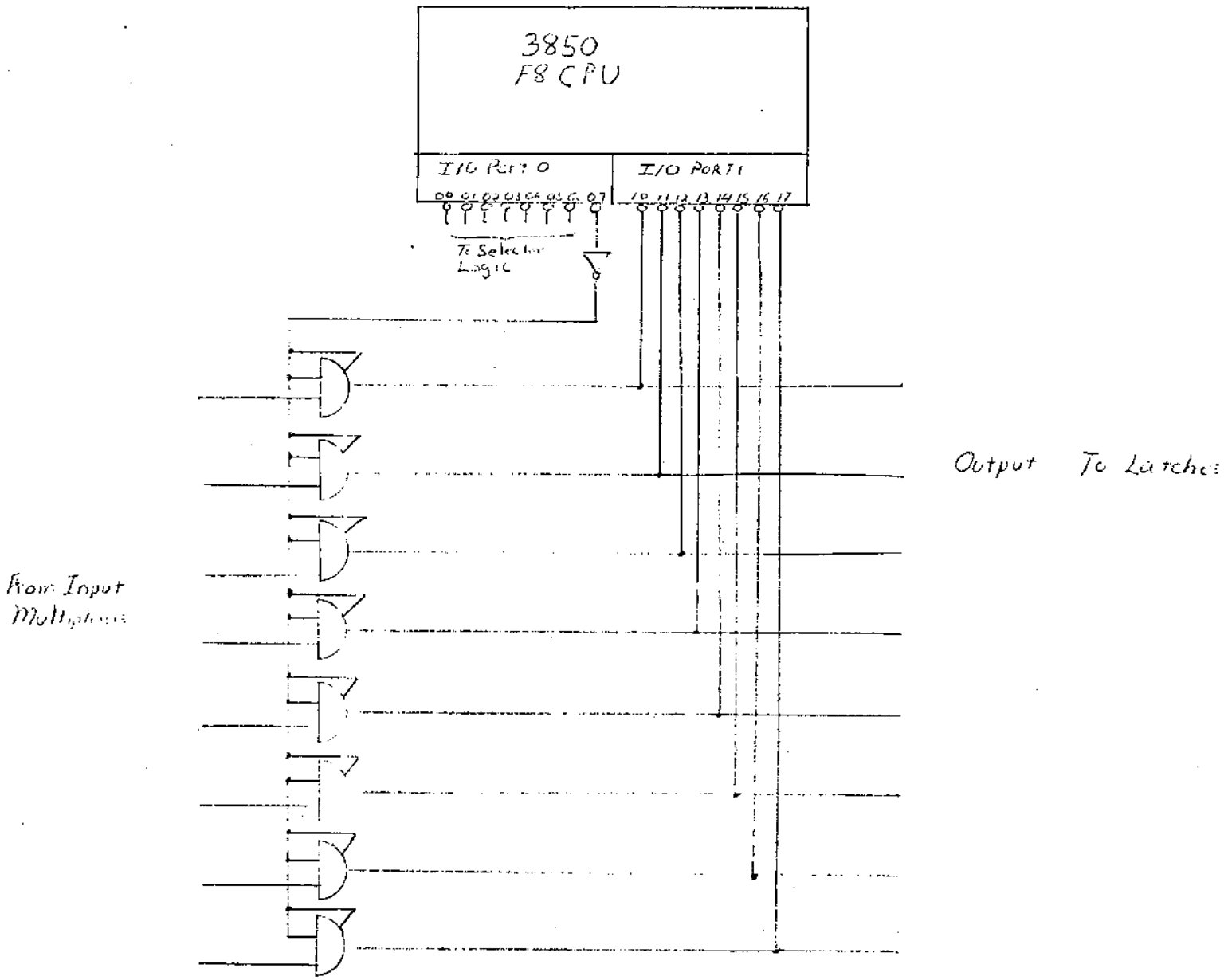
Fig. 1



Expanded Input To F8 I/O Ports

Fig 2

Part Number	Quantity
93150	8
9316	1



Simultaneous I/O Expansion
Fig 3

EXTENDING I/O PORTS FROM THE DATA BUS

Hardware, external to the data bus, may be added to an F8 extending the I/O capability of the microprocessor system. Design of this external logic requires an understanding of the input and output instructions. The advantage of this technique is an economical increase of I/O capability without additional programming.

Input and Output Instructions

There are two F8 input instructions, IN and INS, and two F8 output instructions, OUT and OUTS. Each instruction requires three machine cycles to execute. The IN and OUT instructions are each two bytes long. The first byte is an 8-bit op code and the second byte, the operand, is an 8-bit port address. The INS and OUTS instructions are only one byte long. The first four bits are used for the op code and the remianing bits form the lower half byte of the port address. (The four high order bits of the port address for these instructions are set to be zero). Thus, the IN and OUT instructions may select one of 256 port address while the INS and OUTS only selects one of 16 ports.

All four I/O instructions require three machine cycles for execution. During the first cycle, the port address is available on the data bus. The second cycle is used to pass data to or from the accumulator while the third and final cycle is simply the next instruction fetch. The CPU dictates the state of the peripheral F8 circuits (F8 ROM, F8 MI, and F8 DMA) via five control lines. During an I/O transfer, the second cycle, these five lines will be at the following levels;

	Control Line Levels	
Control Line	OUTS or OUT	INS or IN
ROMC4	1	1
ROMC3	1	1
ROMC2	0	0
ROMC1	1	1
ROMC0	0	1

For an output instruction the data will be passed from the accumulator to the I/O port, while an input instruction transfers data to the accumulator from the port.

I/O Logic

A block diagram of the logic necessary for additional I/O ports is shown below. This circuit is connected to the F8 data bus.

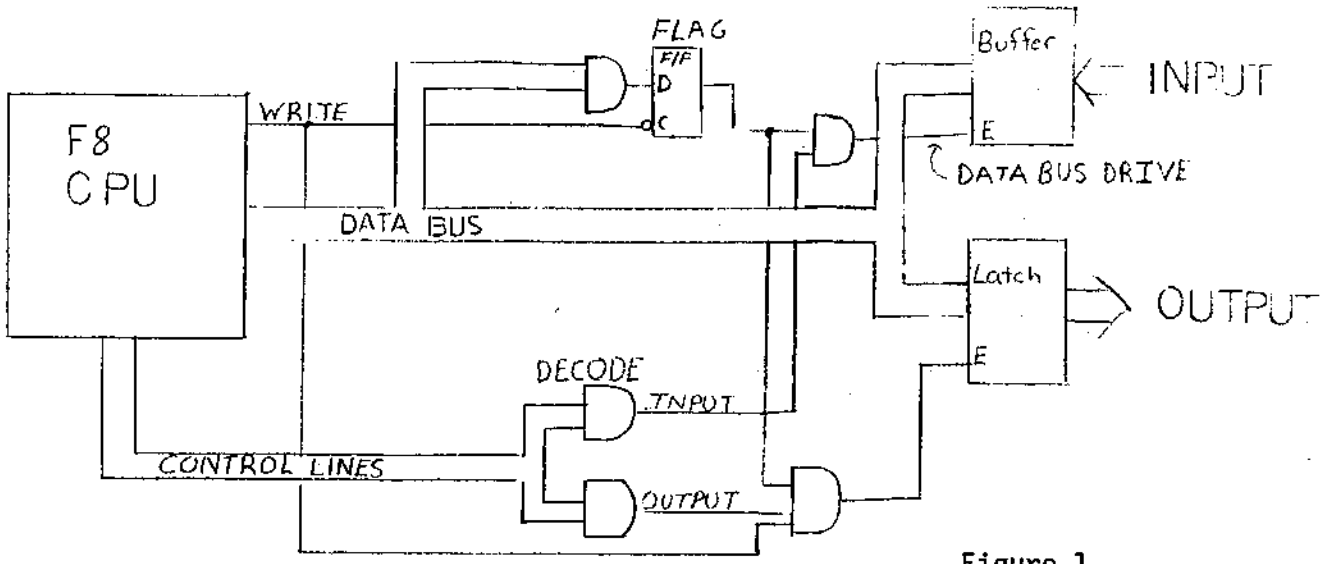
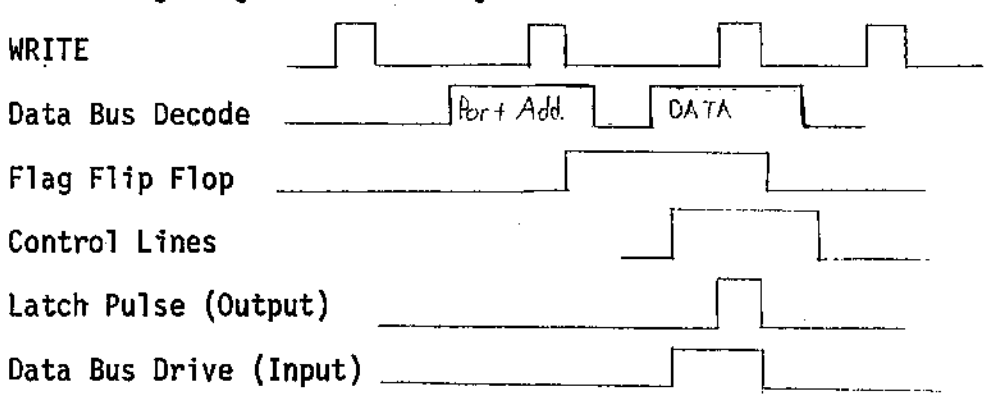


Figure 1

Each new port in the system must be assigned a port number for the instructions to reference. This circuit must decode the data bus for the specific port address. If a match is found, it will flag the match in a flip flop. Next, the CPU control lines are decoded to the Input and Output states. If an input instruction is found, the transfer buffer connected to the data bus is opened. If an output instruction is detected, the external latch is enabled to retain the content of the data bus. If neither instruction is detected,no action is performed. The WRITE pulse, generated by the CPU on the control lines, performs the system timing. This pulse identifies the end of machine cycles as well as time slots when valid data is available on the Data Bus and control lines.

The Timing diagram for this system is shown below.



The latch pulse is used to open the external latches while the data bus drive line allows the input source data to be put onto the data bus.

A very important consideration in this system must be given to the load that the external logic places on the CPU generated signals. Within an F8 system, the data bus lines, the CPU control lines, and the timing signals are primarily intended as intrasystem; i.e. they originate from an F8-MOS circuit and drive one or more F8-MOS circuits. As such, they have very little drive capability. For this reason, it is recommended that CMOS circuits be used to interface with the F8 signals and the external logic. In addition, a CMOS three state circuit is required for driving the data bus.

Example

As an illustration of this technique, Figure 2 is an example of an F8 system with an expansion to eight additional I/O ports. They are assigned port addresses H'F8' through H'FF'. Note the use of CMOS circuits buffering the CPU signals.

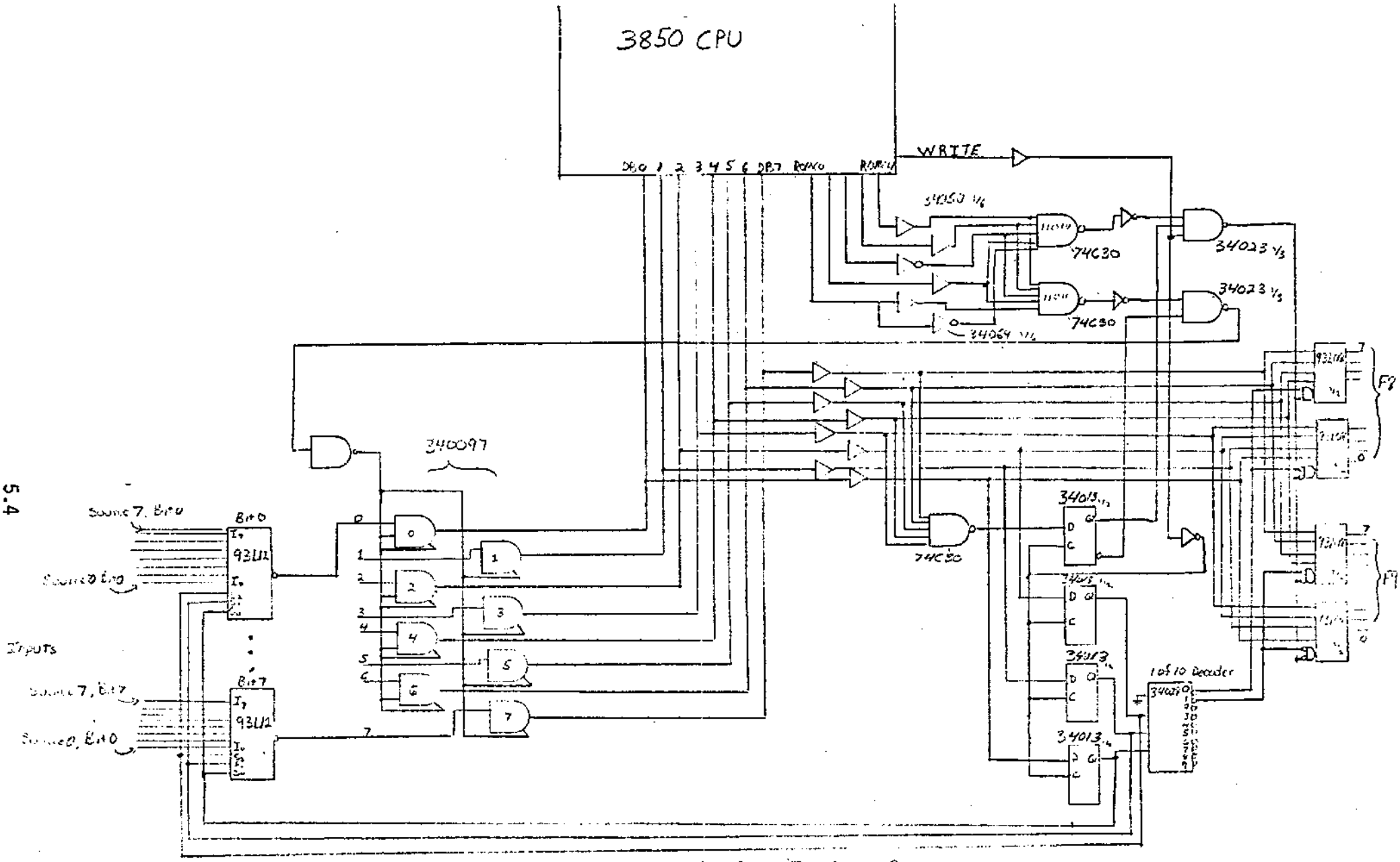
Since eight additional ports are required, it is necessary not only to flag an I/O port address match from the data bus, but to retain the selected port number. This is implemented using four flip flops; one retains an address match while the other three select the port.

The eight input sources to this system are fed into eight multiplexers. The select lines of these multiplexers are obtained from the three select flip flops. The outputs of these devices are connected to CMOS three state non-inverting gates. The drive control signal for these gates originates from the decoding logic.

The eight output latches receive their input from the data bus through CMOS buffers. The enable lines are controlled by the decoding logic and the WRITE pulse. To select one of eight latches, the outputs of the three select flip flops are fed into a decoder. Eight outputs of this decoder are directly connected to the enable line of the latch.

Conclusion

It is possible to add external circuits to an F8 system to increase the number of available I/O ports. The example given requires 29 additional circuits to increase the number of ports by eight. Twelve of these circuits perform a control function while nine are required for eight output latches and eight are necessary for input sources. The advantage of this approach is that it requires no additional programming steps.



Expanding I/O From F8 Data Bus

Figure 2

EXPANDING INPUT SOURCES THROUGH MEMORY

Abstract

Data may be entered into an F8 system using memory locations. A minimum number of external components are required to decode the address lines, select an input source, and drive the data bus. This approach increases the number of instructions that may be used to transfer data into the accumulator while, simultaneously, drastically reducing the programming required in certain applications.

Description

The accompanying figure is an F8 system with the number of input sources expanded to 16. The 16 8-bit inputs are fed into eight multiplexers. Address lines, originating from the Memory Interface circuit, are decoded to memory locations FFF0 through FFFF. (Of course, these lines may be used for other parts of the system not shown here). The lower four bits of the address are fed into the select lines of the multiplexers; thus, they may be used to select one of sixteen input sources. The decoding of the address lines enables the three state gates to drive the F8 data bus.

This method of input expansion offers a major advantage to the user. Instead of only having one instruction, IN or INS, to transfer data into the accumulator, a much larger set of instructions, the Memory Reference Instructions, may be used. Thus, the input data may be brought in and an arithmetic operation may be performed using only one instruction. For example, the XM (Exclusive OR with Memory) instruction may be used. The following operation will take place.

$$ACC \leftarrow (ACC) \oplus \text{INPUT DATA}$$

This operation would have required two instructions if, instead, an I/O port was used.

Example

Suppose a system has 128 bits to poll. Each bit may be the service flag bit of a peripheral in the system. A logic zero in the flag means that the unit does not need attention while a one is a request for action. A routine may be

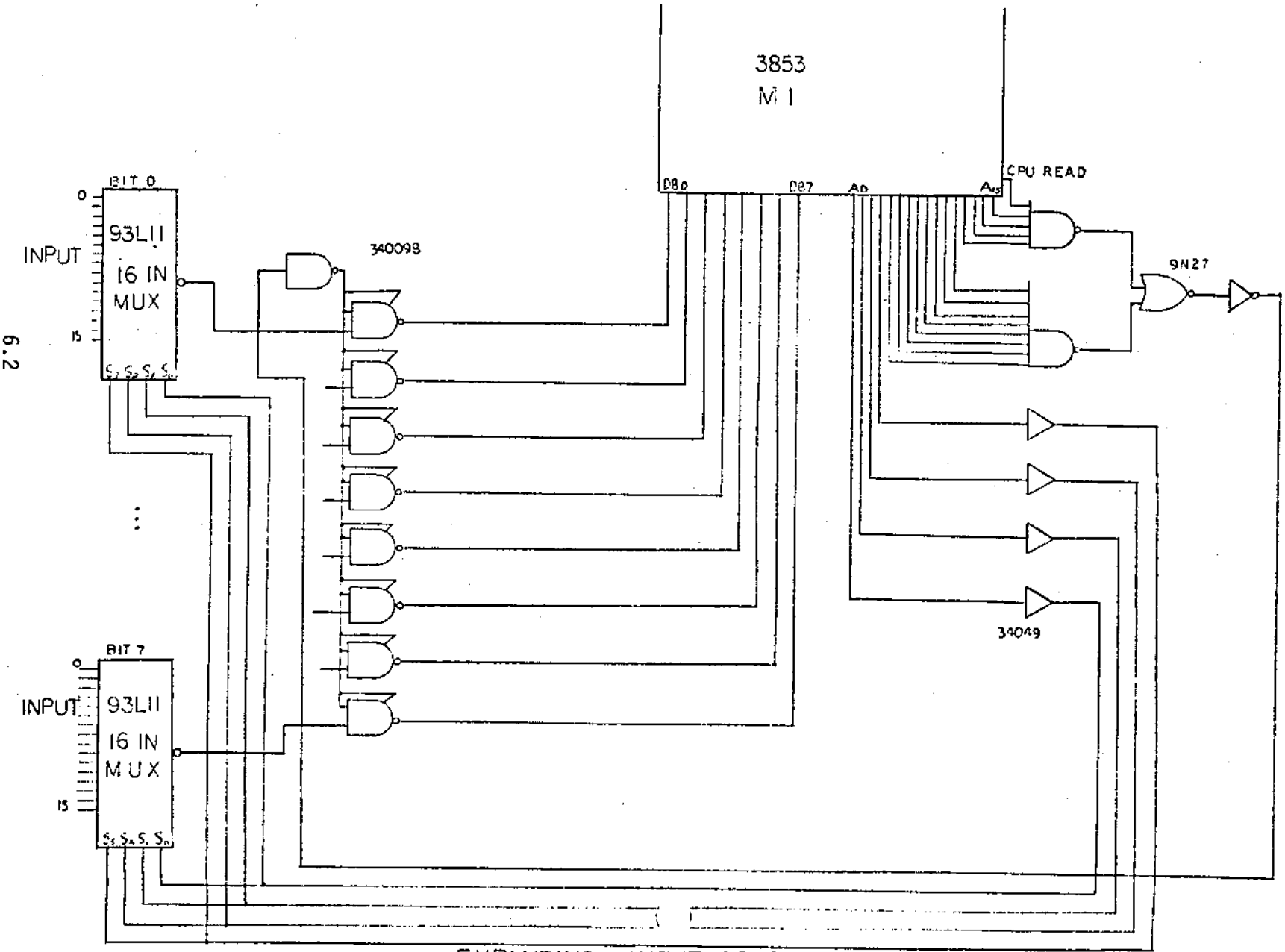
written to accomplish this.

	Operation	Operand	Comment
	DCI	H'FFF0'	Load the data counter with the address of the first input source
	LIS	H'F'	Put 1111 in the accumulator
	LR	1, A	Store 1111 in a counter
	LIS	0	Clear the accumulator
TOP	CM		Compare Input Source to zero
	BNZ	EXIT	Branch if Input is not zero
	DS	1	Decrement the counter
	BNZ	TOP	Branch if counter is not zero

This routine will first load the data counter with the input address. This is a 16-bit address because the memory space is being used for the data. Scratchpad register one is the loop counter. Next, the program clears the accumulator and, using the CM instruction, compares the input against zero. If the result is non-zero, a jump to EXIT will occur. For a zero result the counter will be decremented. The counter serves two purposes. First, it signals the end of the loop, when all sources have been polled. The counter also identifies which input byte was non-zero.

Conclusion

Placing input sources in memory provides a simple scheme for performing certain input tasks. Polling many inputs is one of these. The memory reference instructions offer much flexibility during input operations. This concept may also be expanded to output latches. In a similar fashion, the address lines may be decoded to load a latch using a Memory Reference instruction.



PARTS

93L11	8
340098	2
34049	1
9N27	1

EXPANDING INPUT SOURCES THRU MEMORY

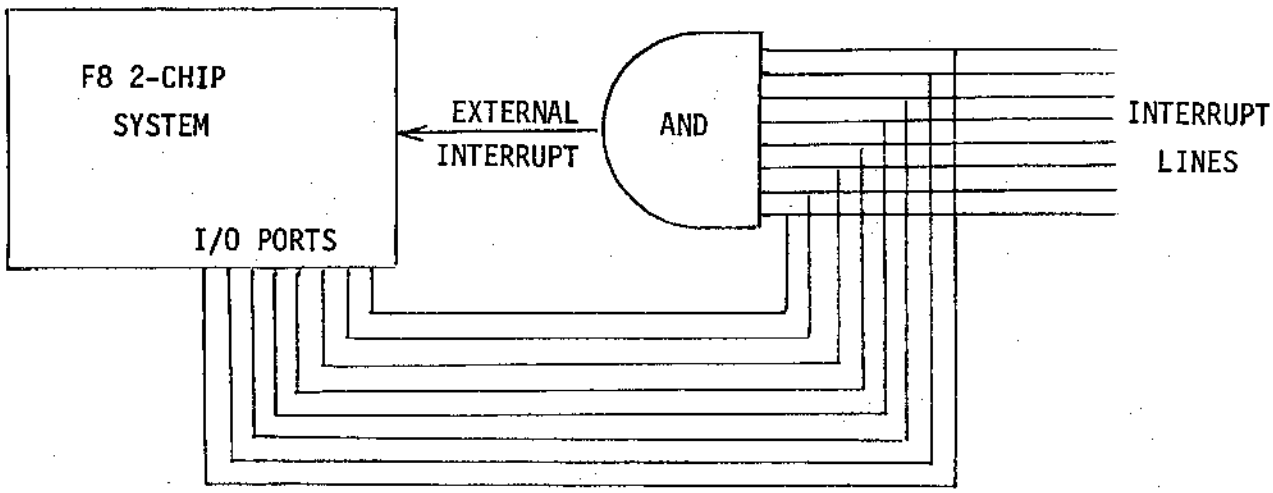
EXTENSION OF EXTERNAL PRIORITY INTERRUPT LEVELS IN A F8 2 CHIP CPU-ROM CONFIGURATION

Abstract

The F8 2 chip (CPU-ROM) system has only 1 level of external interrupt. However, it takes only 1 additional gate to extend the levels to as many as up to 32 levels. The following description, however, is confined to an extension of up to 8 levels.

Description

Fig. 1 describes such a configuration

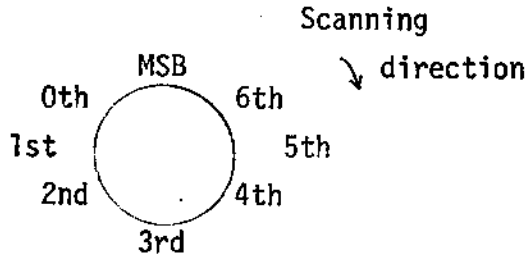


The Interrupt Line and the I/O Port Lines are all negative signals. The 'AND' gate is actually an 'OR' function of the negative signals.

Fig. 1

All the interrupt lines are 'AND'ed together to give a common interrupt signal to the ROM chip. However, each interrupt line is also tied to a line of an I/O port according to a priority basis. (e.g. the highest priority on the MSB line to the least priority on the LSB line).

Once the system is interrupted by 1 of the eight lines, the system will go into an interrupt scanning routine which will scan for the active line on the I/O port from left to right, the result from that interrupt scan will be a code indicating which interrupt is active and ought to be serviced.



Scanning routine

```
Scan      LIS     8       Load a count of A
           LR     0,A     Store in Register 0
          INS     PORT    Get the interrupt information
           LR     1,A     Store in Register 1
Loop       AS     1       Add REG 1
           BC     OUT     MSB is a '1'
           LR     1,A     Put the left shifted result
           DS     0       back to R1
           BR     LOOP    Update the code
                          Do it again.

OUT . . . . . . . . . . . . . .
```

With the above scheme, a multi-level priority interrupt scheme is constructed. However, the user should also be aware of the timing of the interrupts responses since the scanning also takes time.

DOUBLE PRECISION BINARY ADD/SUBTRACT SUBROUTINE

Purpose

This subroutine will perform a 16-bit binary addition or subtraction and return to the user with the status of the 16-bit operation and the result of the operation.

Calling Sequence

Scratch Regs: 1,0 Contain AUGEND/SUBTRAHEND

3,2 Contain ADDEND/MINUEND

(bit 7 of SR1 and SR3 is the most significant bit and sign bit of the 16-bit numbers). SR7 = 0 for add, SR7 ≠ for subtract.

PI DBA (Calls subroutine to perform add or subtract

Upon Exit

The sum or difference is contained in SR3, SR2 with bit 7 of SR3 being the most significant bit and the sign bit.

The proper status is set in the W-Reg for the 16-bit operation.

Operation Performed

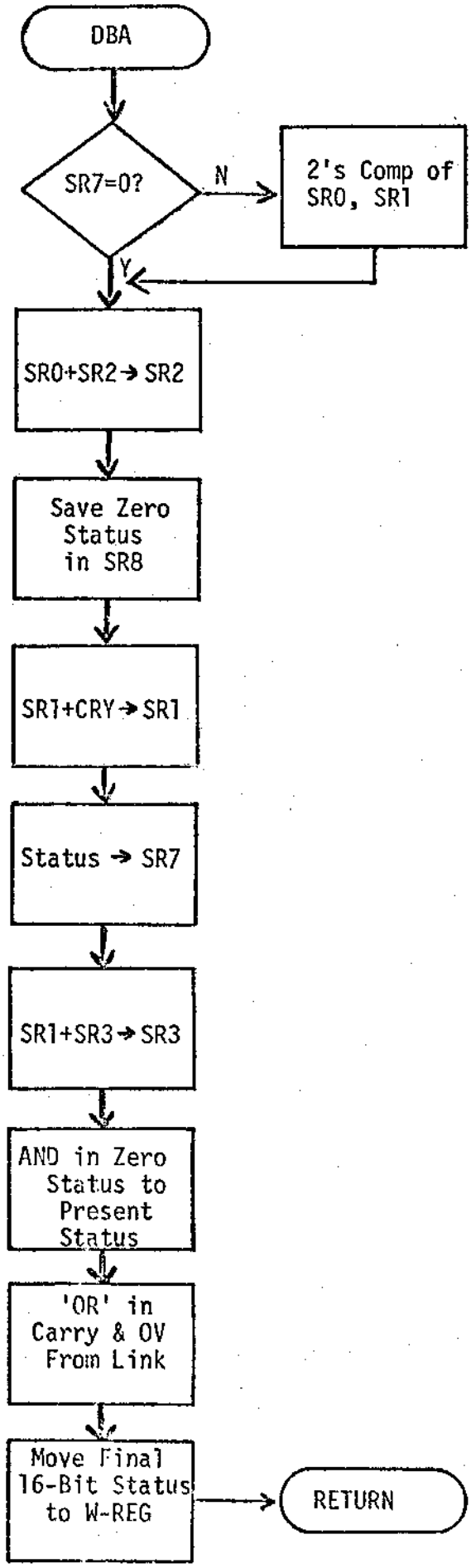
(SR3, SR2) ± (SR1, SR0) ⟶ (SR3, SR2)

Registers Used

ACC, W, J, SR0-SR3, SR7-SR8

Calls to other Subroutines

None


DBA
SR7=0?
N
2's Comp of SR0, SR1
Y
SR0+SR2 → SR2
Save Zero Status in SR8
SR1+CRY → SR1
Status → SR7
SR1+SR3 → SR3
AND in Zero Status to Present Status
'OR' in Carry & OV From Link
Move Final 16-Bit Status to W-REG
RETURN

```
LOC   OBJECT CODE      STMT   SOURCE STATEMENT

                              * DOUBLE PRECISION BINARY ADD (16 BIT) OR
                              * SUBTRACT SUBROUTINE.
                              *   UPON ENTRY:
                              *     IF SR7 = 0 ADD, OTHERWISE SUBTRACT.
                              *     AUGEND = SCRATCH REGS 0 AND 1(1 MSB).
                              *     ADDEND = SCRATCH REGS 2 AND 3(3 MSB).
                              *     (THESE DOUBLE REGISTERS ARE TREATED AS
                              *     16 BIT SIGNED NUMBERS WITH BIT 7 OF
                              *     REGS 1 AND 3 BEING THE SIGN BIT)
                              *     (IF SUBTRACT, THEN (R2,R3)-(R0,R1)).
                              *   UPON EXIT:
                              *     SUM OR DIFF IS CONTAINED IN R2,R3.
                              *     W-REG CONTAINS  THE STATUS OF THE 16
                              *     BIT ADD.
20D7   70                 126     DBA  LIS  0
20D8   C7                 127          AS   7
20D9   84   B             128          BZ   ADD     IF SR7=0, THEN GO DO ADD.
20DB   40                 129          LR   A,0
20DC   18                 130          COM
20DD   1F                 131          INC
20DE   50                 132          LR   0,A     COMPLEMENT FOR SUBTRACT.
20DF   1E                 133          LR   J,W
20E0   41                 134          LR   A,1
20E1   18                 135          COM
20E2   1D                 136          LR   W,J
20E3   19                 137          LNK
20E4   51                 138          LR   1,A
20E5   40                 139     ADD  LR   A,0     GET LSB OF AUGEND.
20E6   C2                 140          AS   2       ADD LSB OF ADDEND.
20E7   52                 141          LR   2,A     SAVE PARTIAL SUM.
20E8   1E                 142          LR   J,W
20E9   49                 143          LR   A,9     GET STATUS AND OR IN ALL BITS
20EA   22   1B            144          OI   H'1B'   BUT ZERO.
20EC   58                 145          LR   8,A     SAVE IN SR8 FOR CALC OF FINAL.
20ED   1D                 146          LR   W,J     GET ORIG STATUS BACK IN W-REG.
20EE   41                 147          LR   A,1     GET 2ND BYTE OF AUGEND.
```

```
LOC  OBJECT CODE  STMT  SOURCE STATEMENT
20EF   19          148        LNK          ADD 1 IF CARRY FROM PREV ADD.
20F0   1E          149        LR   J,W     SAVE STATUS, MAY BE OV OR CARRY.
20F1   C3          150        AS   3       ADD LSB OF ADDEND.
20F2   53          151        LR   3,A     SAVE IN REG 1.
20F3   49          152        LR   A,9     MOVE STATUS SAVED FROM LINK
20F4   57          153        LR   7,A     ADD TO SR 7.
20F5   1E          154        LR   J,W
20F6   49          155        LR   A,9     GET STATUS FROM FINAL ADD AND
20F7   F8          156        NS   8       'AND' IN ZERO STATUS FROM
20F8   58          157        LR   8,A     FIRST ADD.
20F9   47          158        LR   A,7
20FA   59          159        LR   9,A     MOVE STATUS SAVED FROM LINK
20FB   48          160        LR   A,8     BACK TO W-REG AND 'OR' IN
20FC   1D          161        LR   W,J     OV AND CARRY BITS TO FINAL
20FD   92    4     162        BNC  DBA1    STATUS.
20FF   22    2     163        OI   2
2101   1D          164        LR   W,J
2102   96    3     165   DBA1 BND  DBAX
2104   22    8     166        OI   8
2106   59          167   DBAX LR   9,A     MOVE FINAL STATUS TO W-REG.
2107   1D          168        LR   W,J
2108   1C          169        POP          RETURN.
```

BINARY MULTIPLY SUBROUTINE

Purpose

This subroutine performs a binary multiplication of two 8-bit numbers resulting in a 16-bit product. The multiplication is accomplished by testing bit positions of the multiplier and adding the corresponding value to a partial product (when a bit is set) until all of the bits have been tested.

Calling Sequence

SR1 Contains the multiplier

SR2 Contains the multiplicand

PI BMPY (Calls subroutine to perform the multiplication)

Upon Exit

The product is contained in SR7, SR6, with bit 7 of SR7 being the most significant bit of the 16-bit product.

Operation Performed

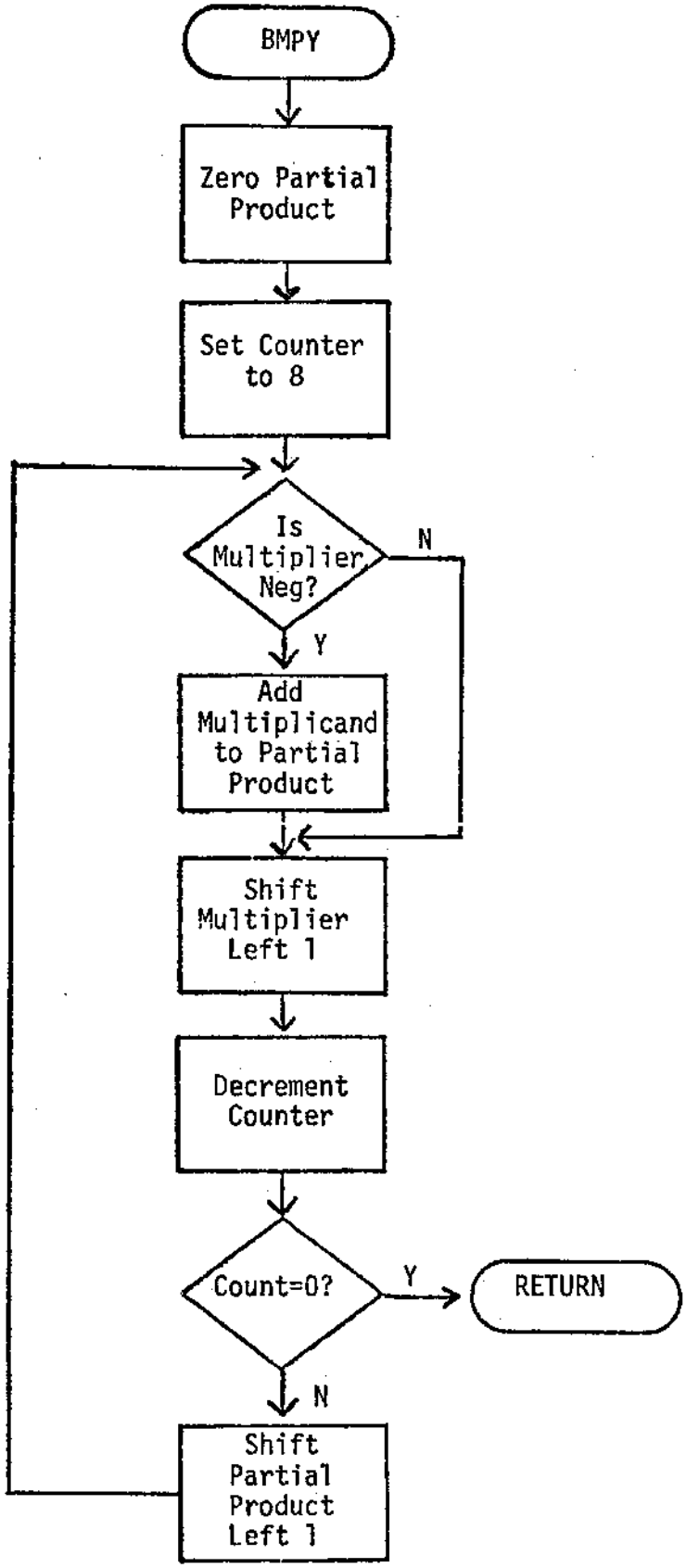
SR1 X SR2 $\longrightarrow$ (SR7, SR6)

Registers Used

ACC, J, W, SR1 - SR2, SR5 - SR7

Calls to Other Subroutines

None


BMPY
Zero Partial Product
Set Counter to 8
Is Multiplier Neg?
N
Y
Add Multiplicand to Partial Product
Shift Multiplier Left 1
Decrement Counter
Count=0?
Y
RETURN
N
Shift Partial Product Left 1

```
LOC   OBJECT CODE        STMT   SOURCE STATEMENT

                         *BINARY MULTIPLY SUBROUTINE

                         *SCRATCH REG 1 CONTAINS MULTIPLIER.

                         *SCRATCH REG 2 CONTAINS MULTIPLICAND.

                         *SCRATCH REGS 6 AND 7 CONTAIN PRODUCT (SR7-MSB).
2125   78                196   BMPY LIS  8       INITIALIZE COUNTER TO 8.
2126   55                197        LR   5,A     INITIALIZE PARTIAL-PRODUCT.
2127   70                198        LIS  0
2128   56                199        LR   6,A
2129   57                200        LR   7,A
212A   70                201   BMP1 LIS  0
212B   C1                202        AS   1       IS SIGN BIT OF MULTIPLIER SET?
212C   81   7            203        BP   BMP2    NO, GO SHIFT PARTIAL PRODUCT.
212E   42                204        LR   A,2
212F   C6                205        AS   6       YES, ADD MULTIPLICAND TO
2130   56                206        LR   6,A     PARTIAL PRODUCT.
2131   47                207        LR   A,7
2132   19                208        LNK
2133   57                209        LR   7,A
2134   41                210   BMP2 LR   A,1
2135   13                211        SL   1       SHIFT THE MULTIPLIER LEFT 1.
2136   51                212        LR   1,A
2137   35                213        DS   5       DECREMENT COUNTER
2138   84   C            214        BZ   BMP3    EXIT IF DONE.
213A   46                215        LR   A,6     SHIFT PARTIAL PRODUCT LEFT 1.
213B   C6                216        AS   6       SHIFT LSB LEFT 1
213C   56                217        LR   6,A
213D   1E                218        LR   J,W     SAVE STATUS FOR CARRY.
213E   47                219        LR   A,7
213F   13                220        SL   1       SHIFT MSB LEFT 1 AND ADD IN
2140   1D                221        LR   W,J     CARRY FROM PREV SHIFT (LSB)
2141   19                222        LNK
2142   57                223        LR   7,A
2143   90   E6           224        BR   BMP1    GO BACK FOR NEXT.
2145   1C                225   BMP3 POP         RETURN TO CALLER.
                         226        END
```

DOUBLE PRECISION BINARY DIVIDE SUBROUTINE

Purpose

This subroutine performs the binary division of a 16-bit unsigned value by another 16-bit unsigned value, and returns with a 16-bit unsigned quotient. The division is performed by repetitively subtracting the divisor from the dividend until a "Carry" is not detected.

Calling Sequence

SR1,-SR0 Contain the divisor (bit 7 of SR1 is most significant bit)

SR3, SR2 Contain the dividend (bit 7 of SR3 is most significant bit)

PI BVD (Calls subroutine to perform the division)

Upon Exit

The quotient is contained in SR6, SR5, with bit 7 of SR6 being the most significant bit of the 16-bit quotient.

Operation Performed

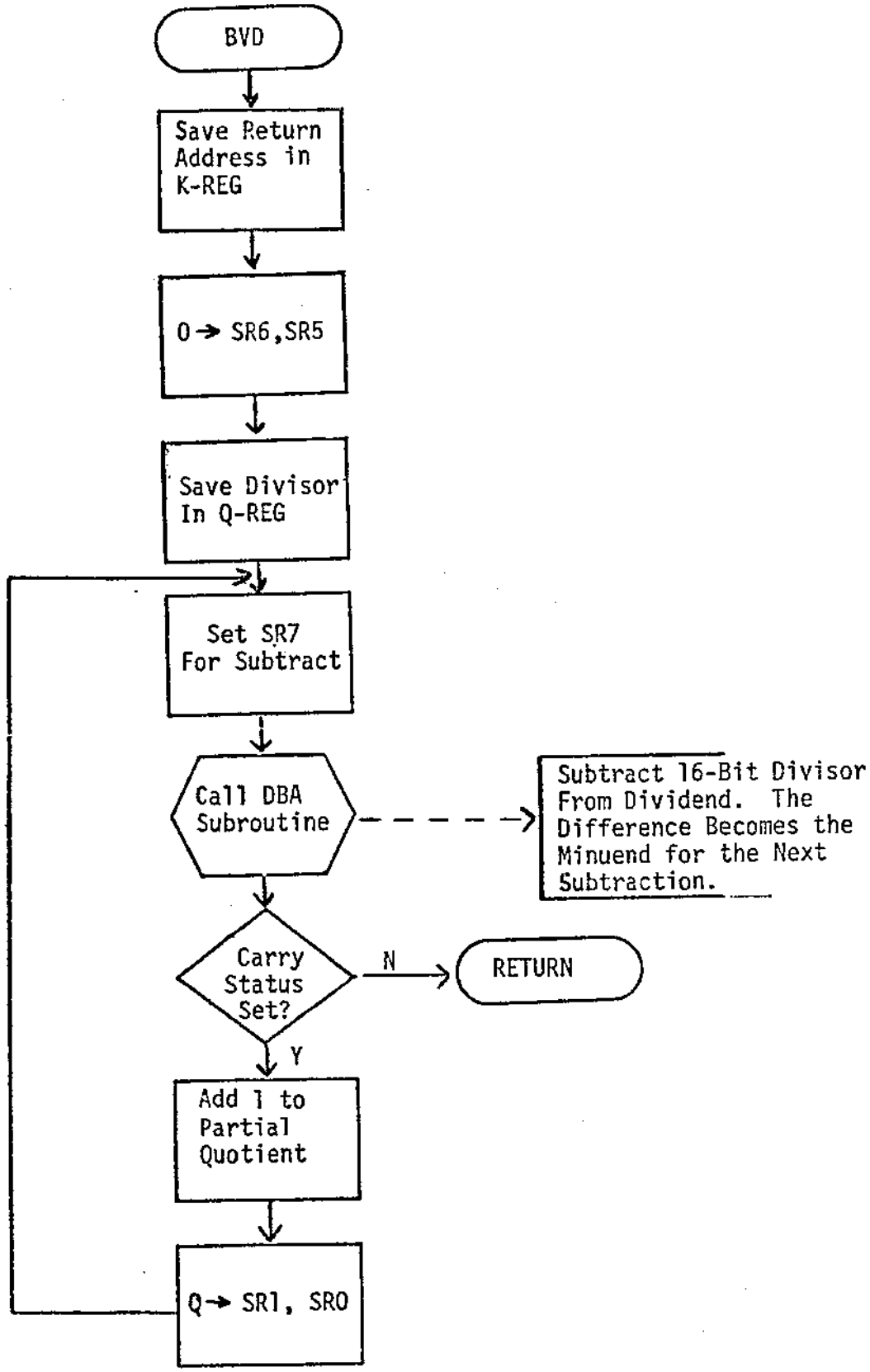
(SR3, SR2) ÷ (SR1, SR0) ⟶ (SR6, SR5)

Registers Used

ACC, K, W, SR0 - SR3, SR5 - SR7

Calls to Other Subroutines

DBA is called to perform the double precision subtraction.


BVD
Save Return Address in K-REG
0 → SR6,SR5
Save Divisor In Q-REG
Set SR7 For Subtract
Call DBA Subroutine
Subtract 16-Bit Divisor From Dividend. The Difference Becomes the Minuend for the Next Subtraction.
Carry Status Set?
N
RETURN
Y
Add 1 to Partial Quotient
Q → SR1, SR0

```
LOC   OBJECT CODE        STMT   SOURCE STATEMENT                                    PAGE   6

                                * DOUBLE PRECISION (16-BIT) BINARY DIVIDE
                                *SUBROUTINE.
                                *   UPON ENTRY:
                                *     SR2,SR3 CONTAIN DIVIDEND (SR3-MSB).
                                *     SR0,SR1 CONTAIN DIVISOR (SR1-MSB).
                                *   CALLS TO OTHER SUBROUTINES:
                                *     PI   DBA   (FOR DOUB. PREC. SUB).
                                *   UPON EXIT:
                                *     SR5,SR6 CONTAIN QUOTIENT (SR6 MSB).
2109    8                 171   BVD  LR   K,P     SAVE RETURN ADDR. IN K.
210A   70                 172        LIS  0
210B   55                 173        LR   5,A     ZERO QUOTIENT AREA.
210C   56                 174        LR   6,A
210D   40                 175        LR   A,0
210E    6                 176        LR   QU,A    SAVE DIVISOR IN Q-REG.
210F   41                 177        LR   A,1
2110    7                 178        LR   QL,A
2111   71                 179   BVD1 LIS  1       SUBTRACT DIVISOR FROM DIVIDEND.
2112   57                 180        LR   7,A     GO TO 'DBA' ROUTINE WITH SR7 NOT
2113   28  20  D7         181        PI   DBA     EQU 0 FOR DOUB. PREC SUBTRACT.
2116   92  D              182        BNC  BVD2

                                * QUOTIENT IS COMPLETE WHEN SUBTRACT RETURNS
                                * WITH 'NO CARRY' STATUS.
2118   45                 183        LR   A,5
2119   1F                 184        INC
211A   55                 185        LR   5,A     INCREMENT QUOTIENT.
211B   46                 186        LR   A,6
211C   19                 187        LNK
211D   56                 188        LR   6,A
211E    2                 189        LR   A,QU
211F   50                 190        LR   0,A     RESTORE DIVISOR.
2120    3                 191        LR   A,QL
2121   51                 192        LR   1,A
2122   90  EE             193        BR   BVD1
2124   1C                 194   BVD2 PK           RETURN TO CALLER.
```

DECIMAL ADD/SUBTRACT SUBROUTINE

Abstract

This subroutine will perform a Floating Decimal ADD or SUBTRACT on two 14 Digit Signed Numbers.

Description

The two numbers are stored using one scratchpad location for every 2 BCD digits The sign uses one half byte, scratchpad location. The exponent of the number uses one scratchpad location. The following scratchpad locations are used in the program example:

K REGISTER

Digit No.	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratch Pad location (Octal)	R27		R26		R25		R24		R23		R22		R21		R20	

R27 Upper, Sign of Number "0" Positive
"F" Negative

R27 Lower, overflow digit

R20 to R26 14 BCD digits, R26 upper - Most significant, R20 lower- Least Significant

The exponent of the number is stored in binary scratchpad location R60 (octal)

Binary "0" represent 10^{-1}
Binary "1" represent 10^{0}
Binary "2" represent 10^{1}
Binary "3" represent 10^{2} etc...

A REGISTER

Digit No	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratch Pad location (Octal)	R37		R36		R35		R34		R33		R32		R31		R30	

R37 Upper, Sign of Number

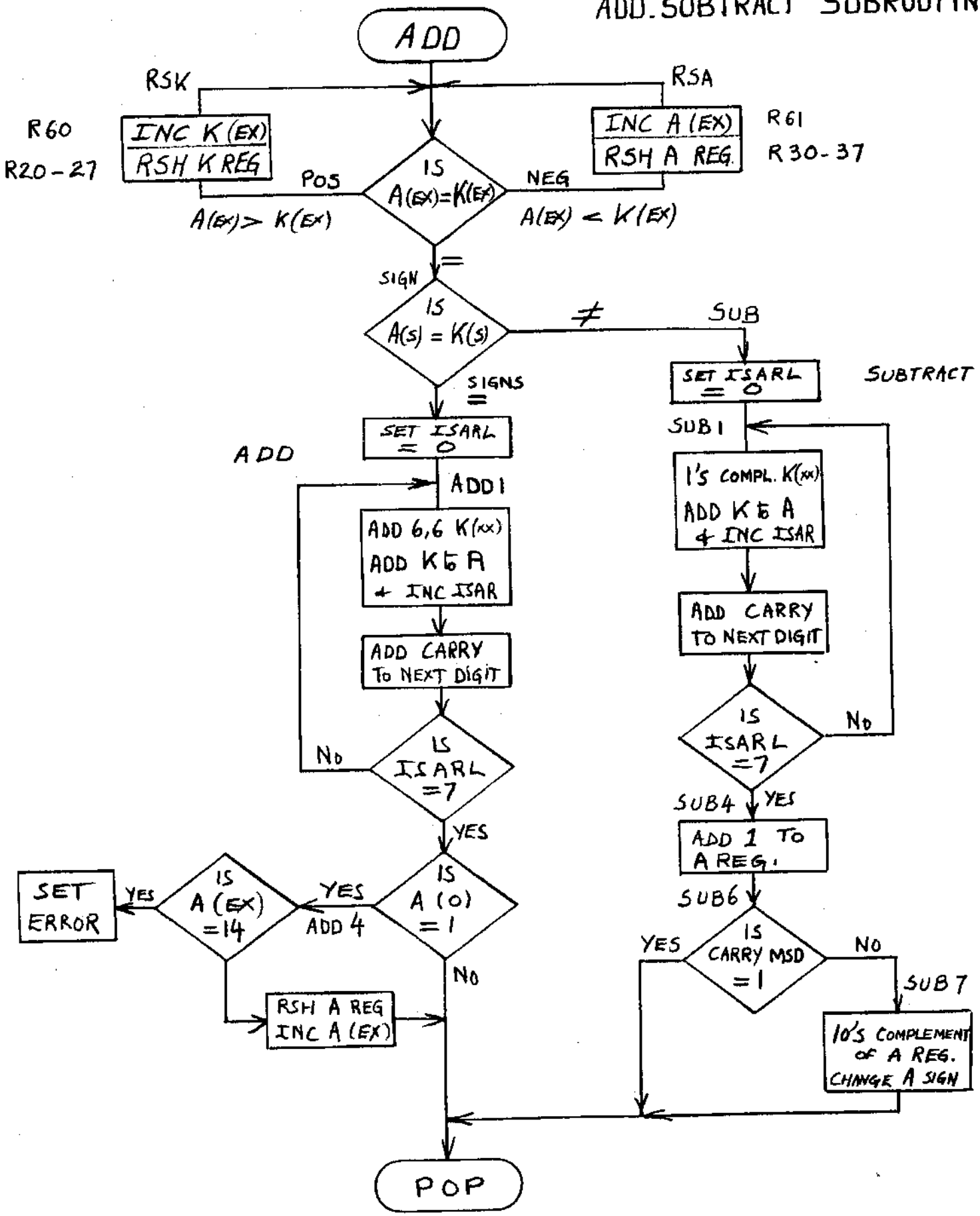
R37 Lower, overflow digit

R30 to R36 14 BCD Digits

The exponent of the number is stored in binary in scratch pad location R61

The number in the K Register is Added to or Subtracted from the number in the A Register. The result is retained in the A Register.

ADD. SUBTRACT SUBROUTINE



```
OBJECT CODE        SOURCE STATEMENT

                   ******* ADD , SUBTACT SUBROUTINE *******
 66                ADD      LISU    6        SET ISAR
 68                         LISL    0            = 60
 4D                         LR      A,I      LOAD K EXP.(R60),INC ISAR
 18                         COM              GET 2'S
 1F                         INC                  COMPLEMENT
 CC                         AS      S        SUB A(EX)-K(EX)
 84  3B                     BZ      SIGN     EQUAL BRANCH
 91  1D                     BM      RSAS     NEGATIVE BRANCH
  8                         LR      K,P      SAVE PC1
 28   2  53                 PI      RSK      GO TO RIGHT SHIFT K(REG)
  9                         LR      P,K      RETURN PC1
 66                         LISU    6        SET ISAR
 68                         LISL    0           =60
 4C                         LR      A,S      LOAD K(EX)
 1F                         INC              INC.K(EX)
 5C                         LR      S,A      STORE K(EX)IN R60
 90  EB                     BR      ADD
 62                RSK      LISU    2        SET ISAR
 68                         LISL    0           =20
 4C                RSK1     LR      A,S      LOAD FROM R(XX)
 14                         SR      4        SHIFT DIGIT RIGHT
 5D                         LR      I,A      STORE IN R(XX),INC ISAR
 4E                         LR      A,D       LOAD FROM R(XX)+1,DEC ISAR
 15                         SL      4        SHIFT DIGIT LEFT
 EC                         XS      S        COMBINE TWO DIGITS
 5D                         LR      I,A      STORE IN R(XX),INC ISAR
 8F  F8                     BR7     RSK1
 20  F0                     LI      H'F0'    CLEAR
 FC                         NS      S           K(0)
 5C                         LR      S,A          R27 LOWER
 1C                         POP
  8                RSAS     LR      K,P      SAVE PC1
 28   2  6F                 PI      RSA      GO TO RIGHT SHIFT A(REG)
  9                         LR      P,K      RETURN PC1
 66                         LISU    6        SET ISAR
 69                         LISL    1           =61
 4C                         LR      A,S      LOAD AND
 1F                         INC                 INC A(EX)
 5C                         LR      S,A      STORE IN R61
 90  CF                     BR      ADD
 63                RSA      LISU    3        SET ISAR
 68                         LISL    0           =30
 4C                RSA1     LR      A,S      RIGHT
 14                         SR      4            SHIFT
 5D                         LR      I,A           A REG.
 4E                         LR      A,D             R30,36
 15                         SL      4                 ONE
 EC                         XS      S                  DIGIT
 5D                         LR      I,A
 8F  F8                     BR7     RSA1
 20  F0                     LI      H'F0'    CLEAR
 FC                         NS      S           A(0)
 5C                         LR      S,A          R37 LOWER
 1C                         POP
 63                SIGN     LISU    3        SET ISAR
 6F                         LISL    7           =37
 20  F0                     LI      H'F0'    ISOLATE UPPER..
 FC                         NS      S        ..DIGIT,A(SIGN) R37
 62                         LISU    2             AND
```

```
OBJECT CODE          SOURCE STATEMENT

 EC                          XS      S        COMPARE WITH K(SIGN) R27
 94  23                      BNZ     SUB      SIGNS DIFFERENT BRANCH
 68                          LISL    0        SIGNS EQUAL ADD
 62                  ADD1    LISU    2
 20  66                      LI      H'66'    BEGIN DECIMAL ADD
 CC                          AS      S        ADD 6,6 TO K DIGIT,R(XX)
 63                          LISU    3
 DC                          ASD     S        ADD DECIMAL K TO A
 5D                          LR      I,A      STORE SUM IN A,INC ISARL
 4C                          LR      A,S      ADD CARRY
 19                          LNK                    IF PRESENT
 5C                          LR      S,A            TO NEXT DIGIT
 8F  F5                      BR7     ADD1
 7F                          LIS     H'F'     ISOLATE LOWER DIGIT
 FC                          NS      S        OF R37 , A(0)
 94   2                      BNZ     ADD4     A(0) =1 BRANCH
 1C                          POP              RETURN
 66                  ADD4    LISU    6        SET ISAR
 69                          LISL    1            = 61
 20   E                      LI      D'14'    LOAD 14
 EC                          XS      S        TEST A(EX), R61
 84  49                      BZ      OVER     A(EX)=14, OVERFLOW
 4C                          LR      A,S      LOAD AND
 1F                          INC                 INCREMENT A(EX)
 5C                          LR      S,A      STORE IN R61
  0                          LR      K,P      SAVE PC1
 28   2  6F                  P1      RSA      GO TO RIGHT SHIFT A(REG)
  9                          LR      P,K      RETURN PC1
 1C                  ADD5    POP              RETURN
 68                  SUB     LISL    0
 62                  SUB1    LISU    2        ISAR =20 TO 27
 4C                          LR      A,S       LOAD K DIGIT R(XX)
 18                          COM              1'S COMPLEMENT OF K DIGIT
 63                          LISU    3
 DC                          ASD     S        SUBTRACT DECIMAL K FROM A
 5D                          LR      I,A      STORE SUM IN A , INC. ISARL
 1E                          LR      J,W      SAVE STATUS OF CARRY
 4C                          LR      A,S      ADD CARRY
 19                          LNK                    IF PRESENT
 5C                          LR      S,A            TO NEXT DIGIT
 8F  F5                      BR7     SUB1
 68                          LISL    0        ISAR = 30
 20  67                      LI      H'67'    ADD 1 TO A 14
 DC                  SUB4    ASD     S        ADD DECIMAL CORRECTION
 5D                          LR      I,A
 4C                          LR      A,S      ADD CARRY
 19                          LNK                   IF PRESENT
 5C                          LR      S,A            TO NEXT DIGIT
 8F   3                      BR7     SUB5     NOT 7 CONTINUE
 90   5                      BR      SUB6     JUMP
```

```
OBJECT CODE          SOURCE STATEMENT

 20  66              SUB5    LI      H'66'
 90  F4                      BR      SUB4
 1D                  SUB6    LR      W,J       LOAD STATUS
 82  1C                      BC      SUB10     CARRY =1 RETURN
 68                          LISL    0         ISAR = 30
 4C                  SUB7    LR      A,S       LOAD A DIGIT FROM R(XX)
 18                          COM               1'S COMPLEMENT OF A DIGIT
 5C                          LR      S,A       STORE BACK DIGIT
 70                          LIS     0         CLEAR ACCUMULATOR
 DC                          ASD     S         SUBTRACT A DIGIT FROM ZERO
 5D                          LR      I,A       STORE A DIGIT ,INC ISARL
 8F  F9                      BR7     SUB7
 68                          LISL    0         ISAR = 30
 20  67                      LI      H'67'     ADD 1 TO A14
 DC                  SUB8    ASD     S         ADD DEC. CORECTION
 5D                          LR      I,A       STORE & INC. ISAR
 4C                          LR      A,S       ADD CARRY
 19                          LNK                  IF PRESENT
 5C                          LR      S,A           TO NEXT DIGIT
 8F   3                      BR7     SUB9      CONTINUE
 90   5                      BR      SUB9A     ISAR=7 JUMP
 20  66                  SUB9    LI      H'66'
 90  F4                          BR      SUB8
 4C                  SUB9A   LR      A,S       SIGN OF A REG (R37)
 18                          COM               CHANGE SIGN OF A
 21  F0                  SUB10   NI      H'F0'     CLEAR A(0) R37 LOWER
 5C                          LR      S,A       STORE BACK DIGIT
 1C                          POP
 29   1  11              OVER    JMP     ERROR     GO TO ERROR
```

DECIMAL MULTIPLY SUBROUTINE

Abstract

This subroutine will perform a Floating Decimal MULTIPLY on two 14 Digit Signed Numbers. The product is also a 14 Digit Signed Number.

Description

The two numbers (multiplicand and multiplier) as well as the product are stored using one scratch pad location for every 2 BCD digits. The sign uses one-half byte or one-half scratch pad location. The exponent of the numbers and product use one scratch pad location each. The following scratch pad locations are used in the program example:

First Number, Multiplicand, F Register

Digit No	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratchpad location (Octal)	R47		R46		R45		R44		R43		R42		R41		R40	

R47 Upper, Sign of Number "0" positive "F" negative

R47 Lower, Overflow digit

R40 to R46, 14 BCD digits, R46 upper - Most Significant, R40 lower - Least Significant

The exponent of the number is stored in binary, in scratchpad location R62 (Octal).

Second Number, Multiplier, W Register

Digit No	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratchpad Location (Octal)	R57		R56		R55		R54		R53		R52		R51		R50	

R57 Upper, Sign of Number

R57 Lower, Overflow Digit

R50 to R56, 14 BCD Digits

The exponent of the number is stored in binary, in scratchpad location R63 (Octal)

Product, K Register

Digit No	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratchpad location (Octal)	R27		R26		R25		R24		R23		R22		R21		R20	

R27 Upper, Sign of Number

R27 Lower, Overflow digit

R20 to R26, 14 BCD digits

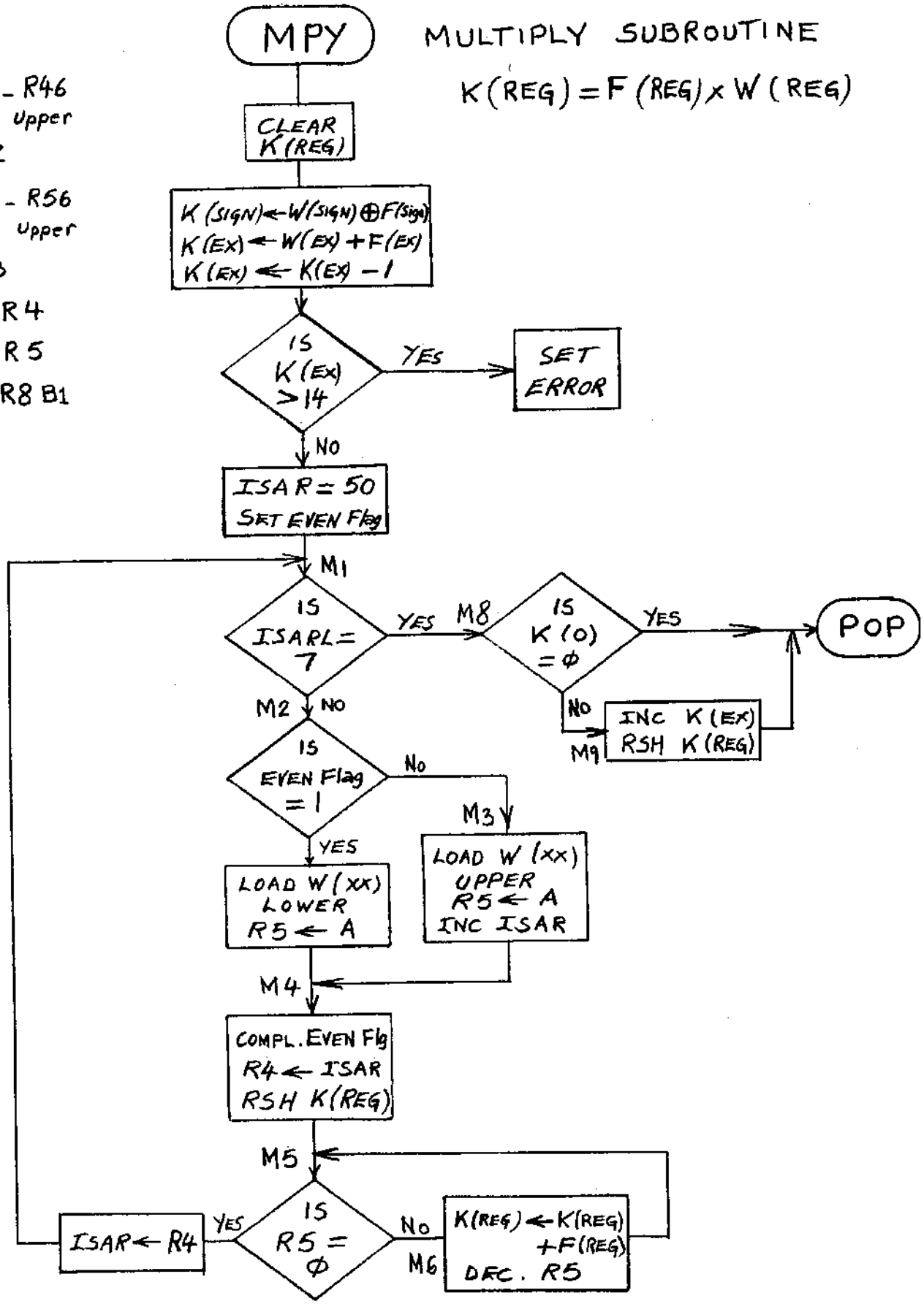
The exponent of the number is stored in binary, in scratchpad location R60.

The number in the F register is multiplied by the number in the W register, the result or Product is stored in the K register with the correct sign and exponent.

F(REG) = R40 - R46
F(SIGN) = R47 upper
F(EX) = R62

W(REG) = R50 - R56
W(SIGN) = R57 upper
W(EX) = R63

TEMP. ISAR = R4
TEMP. DIGIT = R5
EVEN FLAG = R8 B1



```
OBJECT CODE        SOURCE STATEMENT

                   ******* MULTIPLY SUBROUTINE *******
 65                MPY      LISU    5
 6F                         LISL    7          ISAR=57
 20  F0                     LI      H'F0'      ISOLATE SIGN
 FC                         NS      S            OF W(REG)
 64                         LISU    4          ISAR=47
 EC                         XS      S          XOR SIGNS OF F & W (REG)
 62                         LISU    2          ISAR=27
 5E                         LR      D,A        LOAD SIGN OF K(REG)
 70                         LIS     0          CLEAR K(REG)
 5E                         LR      D,A          R26 TO R20
 8F  FE                     BR7     *-1
 66                         LISU    6          ISAR
 6B                         LISL    3            =63
 4E                         LR      A,D        LOAD W(EX)
 CC                         AS      S          ADD F(EX)
 68                         LISL    0          ISAR=60
 5C                         LR      S,A        STORE W(EX)+F(EX) IN K(EX)
 3C                         DS      S          DEC.K(EX)
 24  F1                     AI      H'F1'      ADD MINUS 15
 85  E6                     BT      5,OVER     K(EX) GREATER THAN 14 OVERFLOW
 65                         LISU    5          ISAR =50
 48                         LR      A,8        SET EVEN FLAG
 22   2                     OI      2            R8B1 =1
 58                         LR      8,A            AND STORE
 8F   3            M1       BR7     M2         IS ISAR =7 ?
 90  31                     BR      M8         YES JUMP
 72                M2       LIS     2          IS EVEN FLAG
 F8                         NS      8            = 1 ?
 84   7                     BZ      M3
 4C                         LR      A,S        LOAD W (XX)
 21   F                     NI      H'0F'      ISOLATE LOWER DIGIT
 55                         LR      5,A        STORE IN R5
 90   4                     BR      M4
 4D                M3       LR      A,I        LOAD W(XX)&INC ISAR
 14                         SR      4          SHIFT UPPER DIGIT
 55                         LR      5,A        LOAD MULTIPLIER DIGIT IN R5
 72                M4       LIS     2          COMPLEMENT
 E8                         XS      8            EVEN FLAG
 58                         LR      8,A          R8B1
  A                         LR      A,IS       SAVE ISAR
 54                         LR      4,A          IN R4
  8                         LR      K,P        SAVE PC1
 28   2  53                 PI      RSK        GO TO RIGHT SHIFT K(REG)
```

```
OBJECT CODE        SOURCE STATEMENT

  9                        LR      P,K      RETURN PC1
 20  FF          M5        LI      H'FF'    TEST IF NUMBER
 F5                        NS      5           IN R5 IS ZERO ?
 94   5                    BNZ     M6
 44                        LR      A,4      LOAD R4
  B                        LR      IS,A       IN ISAR
 90  DD                    BR M1
 68              M6        LISL    0        ADD F(REG) TO K(REG)
 64              M7        LISU    4        ISAR = 40
 20  66                    LI      H'66'    BEGIN DECIMAL ADD
 CC                        AS      S        ADD 6,6 TO F(XX)
 62                        LISU    2
 DC                        ASD     S        ADD DECIMAL F(XX) TO K(XX)
 5D                        LR      I,A      STORE SUM IN K, INC ISAR
 4C                        LR      A,S      ADD CARRY
 19                        LNK                 IF PRESENT
 5C                        LR      S,A           TO NEXT DIGIT
 8F  F5                    BR7     M7
 35                        DS      5        DECREMENT NUMBER IN R5
 90  E8                    BR      M5
 62              M8        LISU    2        ISAR =27
 7F                        LIS     H'F'     ISOLATE LOWER DIGIT
 FC                        NS      S           OF R27,K(0)
 94   2                    BNZ     M9       K(0) NOT ZERO , BRANCH
 1C                        POP              K(0) =ZERO , RETURN
 66              M9        LISU    6        SET ISAR
 68                        LISL    0           =60
 4C                        LR      A,S      LOAD AND
 1F                        INC                 INCREMENT K(EX)
 5C                        LR      S,A      STORE IN R 60
  6                        LR      K,P      SAVE PC1
 28   2  53                PI      RSK      GO TO RIGHT SHIFT K(REG)
  9                        LR      P,K      RETURN PC1
 1C                        POP              END OF MPY
```

DECIMAL DIVIDE SUBROUTINE

Abstract

This subroutine will perform a Floating Decimal DIVIDE on two 14 Digit Signed Numbers. The quotient is also a 14 Digit Signed Number.

Description

The two numbers (dividend and divisor) as well as the quotient are stored using one scratchpad location for every 2 BCD digits. The sign uses one-half byte or one-half scratchpad location. The exponent of the numbers and quotient, one scratchpad location each. The following scratchpad locations are used in the program example:

First Number, Divisor, F Register

Digit No	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratchpad location (Octal)	R47		R46		R45		R44		R43		R42		R41		R40	

R47 Upper, Sign of Number "0" positive "F" negative

R47 Lower, Overflow digit

R40 to R46, 14 BCD digits, R46 upper - Most Significant, R40 lower - Least Significant

The exponent of the number is stored in binary, in scratchpad location R62 (Octal).

Second Number, Divident, W Register

Digit No	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratchpad location (Octal)	R57		R56		R55		R54		R53		R52		R51		R50	

R57 Upper, Sign of Number

R57 Lower, Overflow Digit

R50 to R56, 14 BCD Digits

The exponent of the number is stored in binary, in scratchpad location R63 (Octal).

Quotient, K Register

Digit No	Sign	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Scratchpad location (Octal)	R27		R26		R25		R24		R23		R22		R21		R20	

R27 Upper, Sign of Number

R27 Lower, Overflow Digit

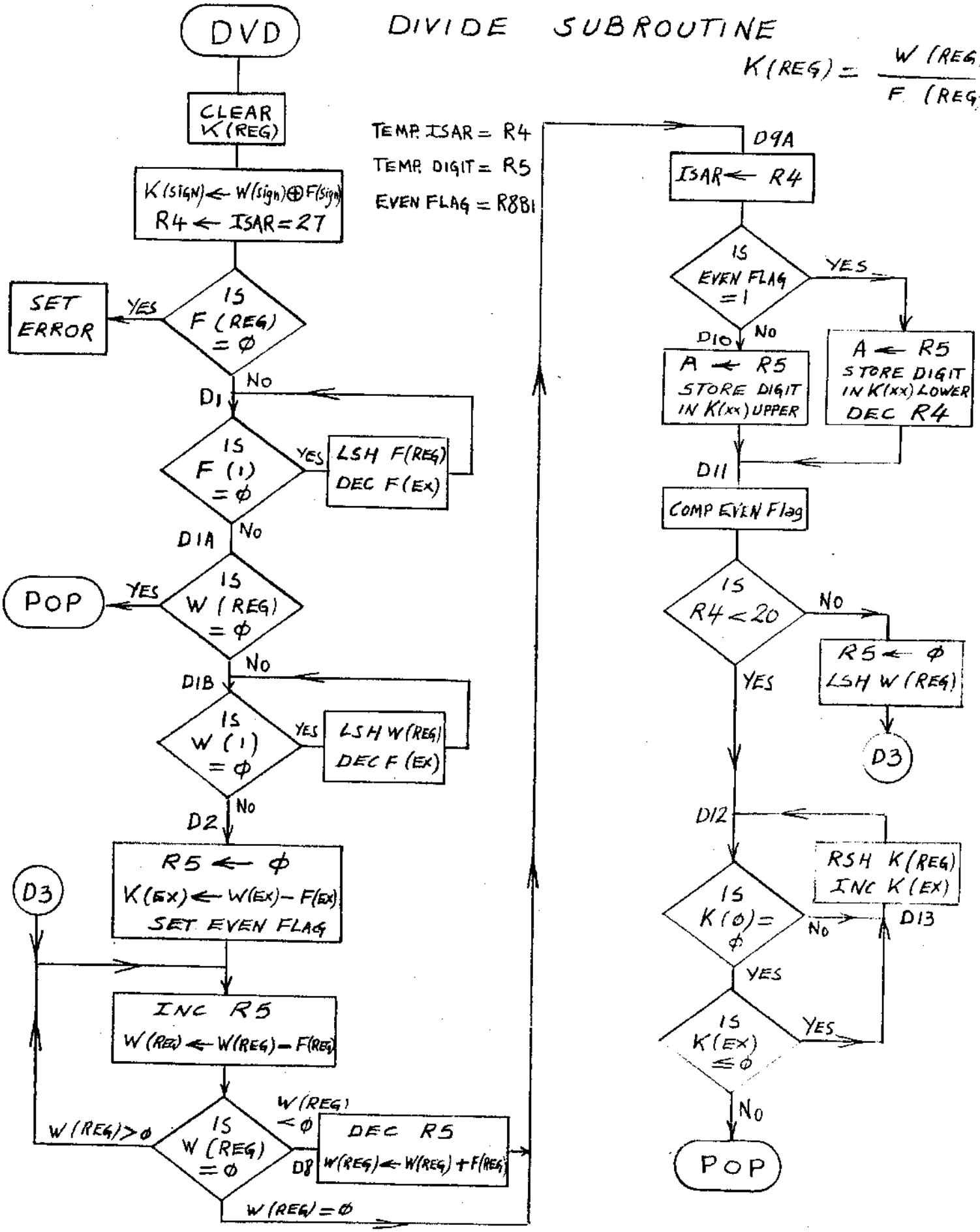
R20 to R26, 14 BCD digits

The exponent of the number is stored in binary, in scratchpad location R60.

The number in the W register is divided by the number in the F register, the result or quotient is stored in the K register with the correct sign and exponent.

DIVIDE SUBROUTINE

$$K(REG) = \frac{W(REG)}{F(REG)}$$



```
                        ******* DIVIDE SUBROUTINE *******
65                      DVD     LISU    5
6F                              LISL    7       ISAR=57
20  F0                          LI      H'F0'   ISOLATE SIGN
FC                              NS      S         OF W (REG)
64                              LISU    4       ISAR=47
EC                              XS      S       XOR SIGNS OF W & F (REG)
62                              LISU    2       ISAR=27
5E                              LR      D,A     LOAD SIGN OF K (REG)
70                              LIS     0       CLEAR K(REG)
5E                              LR      D,A         R26 TO R20
8F  FE                          BR7     *-1
 A                              LR      A,IS    STORE ISAR
54                              LR      4,A         IN R4
64                              LISU    4       SET ISAR
6E                              LISL    6          =46
70                              LIS     0       LOAD ZERO
CE                              AS      D       TEST F(REG)
94   5                          BNZ     D1      NOT ZERO BRANCH
8F  FC                          BR7     *-3
90  83                          BR      OVER    F(REG)=0 ERROR
64                      D1      LISU    4
6E                              LISL    6       ISAR=46
20  F0                          LI      H'F0'   TEST F(1)
FC                              NS      S        IN R46 UPPER
94   F                          BNZ     D1A     IF NOT ZERO JUMP
4D                              LR      A,I     LEFT
14                              SR      4         SHIFT
EC                              XS      S           F(REG)
5E                              LR      D,A            BY
4C                              LR      A,S             ONE
15                              SL      4                 DIGIT
5E                              LR      D,A
8F  F8                          BR7     *-7
66                              LISU    6
6A                              LISL    2       ISAR=62
3C                              DS      S       DECREMENT F(EX)
90  EC                          BR      D1
65                      D1A     LISU    5       SET ISAR
6F                              LISL    7          =57
70                              LIS     0       CLEAR W(0 & SIGN)
```

```
OBJECT CODE          SOURCE STATEMENT

 5E                          LR      D,A          R57
 CE                          AS      D        TEST W(REG)
 94   4                      BNZ     D1B      NOT ZERO BRANCH
 8F  FC                      BR7     *-3
 1C                          POP              W(REG)=0 RETURN
 65                  D1B     LISU    5
 6E                          LISL    6        ISAR=56
 20  F0                      LI      H'F0'    TEST W(1)
 FC                          NS      S          IN R56 UPPER
 94   F                      BNZ     D2       NOT ZERO JUMP
 4D                          LR      A,I      LEFT
 14                          SR      4          SHIFT
 EC                          XS      S            W(REG)
 5E                          LR      D,A            BY
 4C                          LR      A,S              ONE
 15                          SL      4                  DIGIT
 5E                          LR      D,A
 8F  F8                      BR7     *-7
 66                          LISU    6
 6B                          LISL    3        ISAR=63
 3C                          DS      S        DECREMENT W(EX)
 90  EC                      BR      D1B
 70                  D2      LIS     0
 55                          LR      5,A      CLEAR TEMP. QUOTIENT R5
 66                          LISU    6        SET ISAR
 6A                          LISL    2            =62
 4D                          LR      A,I      LOAD F(EX)
 18                          COM
 1F                          INC               2'S COMPLEMENT
 CC                          AS      S        W(EX) - F(EX)
 68                          LISL    0        ISAR= 60
 5C                          LR      S,A      STORE NEW K(EX)
 48                          LR      A,8      SET EVEN FLAG
 22   2                      OI      2           R8B1 =1
 58                          LR      8,A        AND STORE
 45                  D3      LR      A,5
 1F                          INC              INCREMENT
 55                          LR      5,A        TEMP. QUOTIENT IN R5
 68                          LISL    0
 64                  D4      LISU    4        ISAR =40
 4C                          LR      A,S      LOAD F(XX)
 18                          COM              1'S COMPLEMENT
 65                          LISU    5
 DC                          ASD     S        SUBTACT DEC.F(XX) FROM W(XX)
 5D                          LR      I,A      STORE RESULT IN W(XX),INC ISAR
 4C                          LR      A,S      ADD CARRY
 19                          LNK                 IF PRESENT
 5C                          LR      S,A          STORE IN W(XX)+1
 8F  F6                      BR7     D4
 20  FF                      LI      H'FF'    SUBTACT '00' FROM R57
```

```
OBJECT CODE          SOURCE STATEMENT

 DC                          ASD     S
 5C                          LR      S,A        AND STORE
 1E                          LR      J,W      SAVE STATUS OF CARRY
 68                          LISL    0
 20   67                     LI      H'67'    ADD 1 TO W(14)
 DC                  D5      ASD     S        ADD DECIMAL CORECTION
 5D                          LR      I,A        AND STORE
 4C                          LR      A,S
 19                          LNK              ADD CARRY IF PRESENT
 5C                          LR      S,A        AND STORE
 8F    3                     BR7     D6
 90    5                     BR      D7
 20   66                  D6 LI      H'66'
 90   F4                     BR      D5
 20   66                  D7 LI      H'66'    ADD '00' TO
 DC                          ASD     S          R57
 5C                          LR      S,A        AND STORE
 82   18                     BC      D9A      W(REG) =0 BRANCH
 1D                          LR      W,J      LOAD STATUS
 92    3                     BNC     D8       RESULT NEGATIVE ADD F(REG) TO W(RE(
 90   D2                     BR      D3
 35                      D8 DS      5        DEC. TEMP. QUOTIENT
 68                          LISL    0
 64                      D9 LISU    4
 20   66                     LI      H'66'
 CC                          AS      S        ADD 6,6 TO F DIGITS F(XX)
 65                          LISU    5
 DC                          ASD     S        ADD DECIMAL F(XX) TO W(XX)
 5D                          LR      I,A        AND STORE
 4C                          LR      A,S
 19                          LNK              ADD CARRY IF PRESENT
 5C                          LR      S,A        AND STORE
 8F   F5                     BR7     D9
 20   66                     LI      H'66'    ADD '00'
 DC                          ASD     S          TO R57
 5C                          LR      S,A        AND STORE
 44                     D9A  LR      A,4      LOAD R4
  8                          LR      IS,A       IN ISAR
 72                          LIS     2         IS EVEN FLAG
 F8                          NS      8           =1 ?
 84    7                     BZ      D10
 45                          LR      A,5      LOAD QUOTIENT
 EC                          XS      S          IN
 5C                          LR      S,A        K(XX) EVEN
 34                          DS      4        DECREMENT DIGIT COUNTER
 90    4                     BR      D11
 45                     D10  LR      A,5      LOAD QUOTIENT
 15                          SL      4          IN
 5C                          LR      S,A       K(XX) ODD
 72                     D11  LIS     2        COMPLEMENT
```

```
OBJECT CODE          SOURCE STATEMENT

 E8                          XS      8          EVEN FLAG
 58                          LR      8,A        (R8B1)
 7F                          LIS     H'F'     LOAD 15 (OCTAL 17)
 E4                          XS      4
 84  10                      BZ      D12      R4=17 OCTAL , BRANCH
 70                          LIS     0
 55                          LR      5,A      CLEAR TEMP. QUOTIENT
 65                 LSW      LISU    5        SET ISAR
 6E                          LISL    6           =56
 4D                 LSW1     LR      A,I      LEFT
 14                          SR      4          SHIFT
 EC                          XS      5            W REG
 5E                          LR      D,A
 4C                          LR      A,S            BY
 15                          SL      4                ONE
 5E                          LR      D,A                DIGIT
 8F  F8                      BR7     LSW1
 90  9B                      BR      D3
 62                 D12      LISU    2        SET ISAR
 6F                          LISL    7           =27
 7F                          LIS     H'F'
 FC                          NS      5        ISOLATE K(D) , R27 LOWER
 94   A                      BNZ     D13
 66                          LISU    6        SET ISAR
 68                          LISL    0           =60
 70                          LIS     0        TEST
 CC                          AS      5           K(EX) REG
 91   4                      BM      D13      K(EX) NEG. BRANCH
 84   2                      BZ      D13      K(EX) ZERO BRANCH
 1C                          POP
  8                 D13      LR      K,P      SAVE PC1
 28   2  53                  PI      RSK      GO TO RIGHT SHIFT K (EFC)
  9                          LR      P,K      RETURN PC1
 66                          LISU    6        SET ISAR
 68                          LISL    0           =60
 4C                          LR      A,S      LOAD K(EX)
 1F                          INC
 5C                          LR      S,A      INC. K(EX) IN REG
 90  E6                      BR      D12
```

BOOTSTRAP PROGRAM FOR READING PAPERTAPES

Abstract

This application note describes an F8 program that is used to load information from a paper tape into Read/Write memory. The tape may be read from either a serial teletype or a parallel device such as a tape reader; input routines for both type of devices are included. Program size is 250 bytes. Flow chart and operating instructions included.

This application note describes a program that reads a formatted paper tape into an F8 system. The system is assumed to include the F8 MI (Memory Interface) circuit and RAM memory. This program is designed for loading user's programs into an F8 Micromodule system - the program is supplied in two PROMs (256 X 4 each) and is able to load into RAM that occupies addresses H'0800' and H'0BFF' during loading.

The paper tape is typically produced as output from the F8 cross-assembler which resides in a time-sharing service. All information on the tape is in ASCII representation (i.e. the character 'A' is represented by one line of holes 11000y001 where y is the spocket hole). Each block on the tape begins with an X; the X is then followed by 16 characters from the hex set 0 - 9 and A - F. The 16 characters give 8 data bytes to be loaded into RAM memory. The 16 data characters are followed by a one character check sum; the checksum is used to check the validity of the input. The entire block is typically followed by a carriage return and line feed; other comments can also be inserted if "X"s and "S"s are not included. Since the tape is all ASCII, it can be understood when read and printed. Figure 1 is an example.

The checksum is included in each block of data. It consists of a single character. The checksum is calculated by performing a summation of all 16 half bytes of the block, using hex arithematic. The resulting sum is truncated to its least 4 binary digits; the checksum character is the ASCII representation of the resulting hex digit. When the program reads a paper tape it calculates an internally held

```
******************
******************
******************
******************
S0000**********
X71B0206050A015819
X0E2092512800DD207
X82077006900E209CF
X077006B4B5A51381F
X0390FF2A08007054E
X28009B2553843225 5
X2A843B235894F2572
X78555423007C173 5F
X94FA28009B213F24E
XD08203243918 1FC78
X210F84D5A01581070
X2093512800DD90FFF
X28007C24085A28004
X7C581090BCF484B97
XA015810720935128E
X00DD90FF087256287
X0098213F24D082039
X2439155114C757368
X8405415390EA41149
XC3091C0DA591FE7FE
X181F94FEA41813121
X5171B5A581FE7085A
X411C7952A091FE40E
X189001231F94FBA01
X91F3218DC13291115
X8402125140 18BBBB0
XBB281F94FAA090EB5
XA04181EF1C7B52702
XB04018B3BBBB281F7
X94FA32840C4121D1C
XB041122480519 0EA8
X1C000000000000000D
*********************
*********************
*********************
*********************
```

LEADER

Set Load Address for H'0000'

"X" at start

16 characters represent 8 bytes

Checksum character

Data Lines

Marks end of tape

Figure 1

checksum using these same rules. If the internal checksum does not match the checksum read from the tape, the tape is immediatly stopped--which will be at the end of the bad block.

The paper tape format also includes provisions for setting an address that is to be used for loading data blocks that follow. The format is "SXXXX" where XXXX represents the 4 digit hex load address. There may be more than one load address on a tape--additional load addresses may be inserted between data blocks.

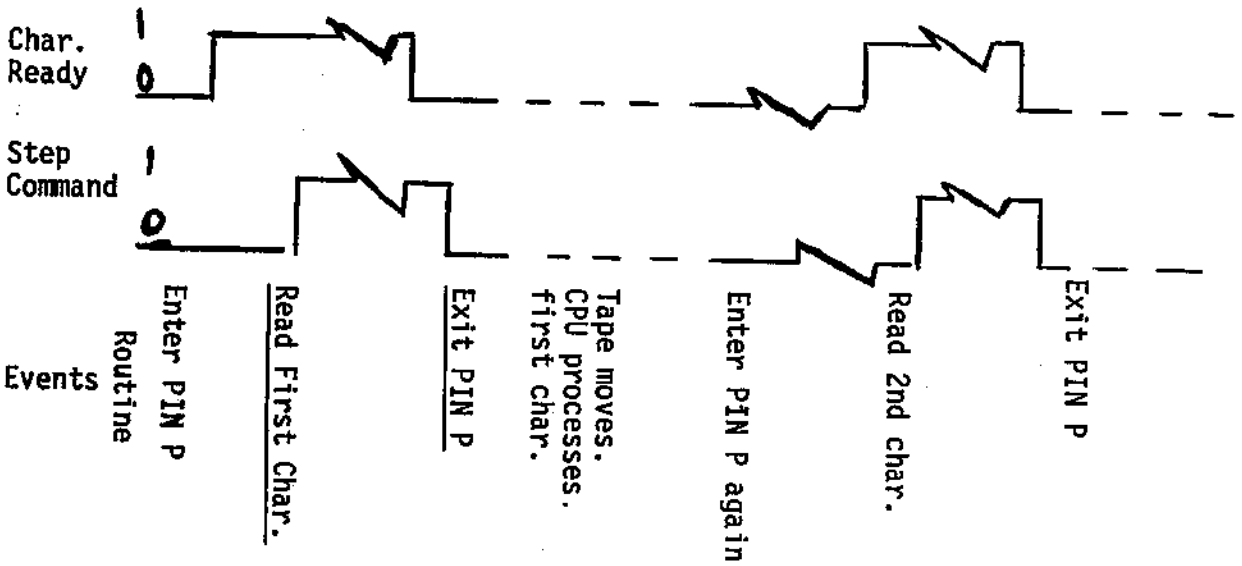
The paper tape may be read from either a serial or parallel device. I/O Port Ø is used for the serial device. If bit 3 of Port Ø is strapped to ground, the program will assume that a serial device is present and that input will be from it. The program sends READER ON, READER OFF control characters to control tape start and stop.

The serial routines use the 11 bit asynchronous format that is standard on teletypes; the 11 bits are 1 start, 8 data,and 2 stop bits. Parity is not checked on input. The teletype routines use software delay loops to set the character rate. The delay loops are set for characters/sec--30 characters/second on a 2usec system; however, the character rate is easily adjusted by changing the number that is loaded into the register "BAUD" at the start of the program.

The parallel input routine uses a generalized "handshaking" protocol that applies to many devices.

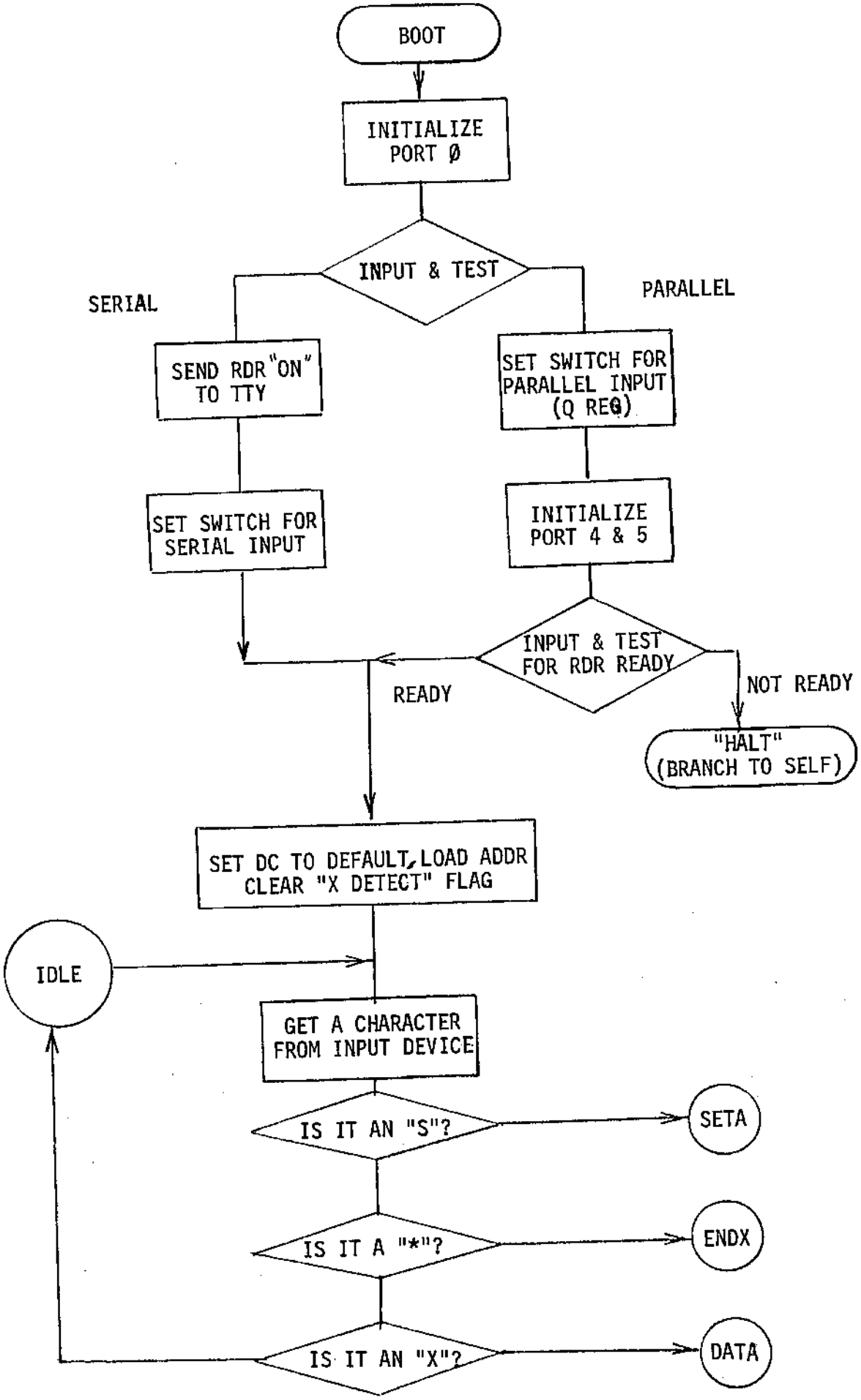
The program waits for the device to indicate ready before reading a character; the device waits for a command from the program before starting to move the tape to the character. Thus slow and fast readers can be used with equal ease. The parallel input could even accommodate a FIFO; the only requirement is that the device be able to operate in the step mode and that the character ready stay low after the step command is received for at least the minimum width of that command: 24usec with

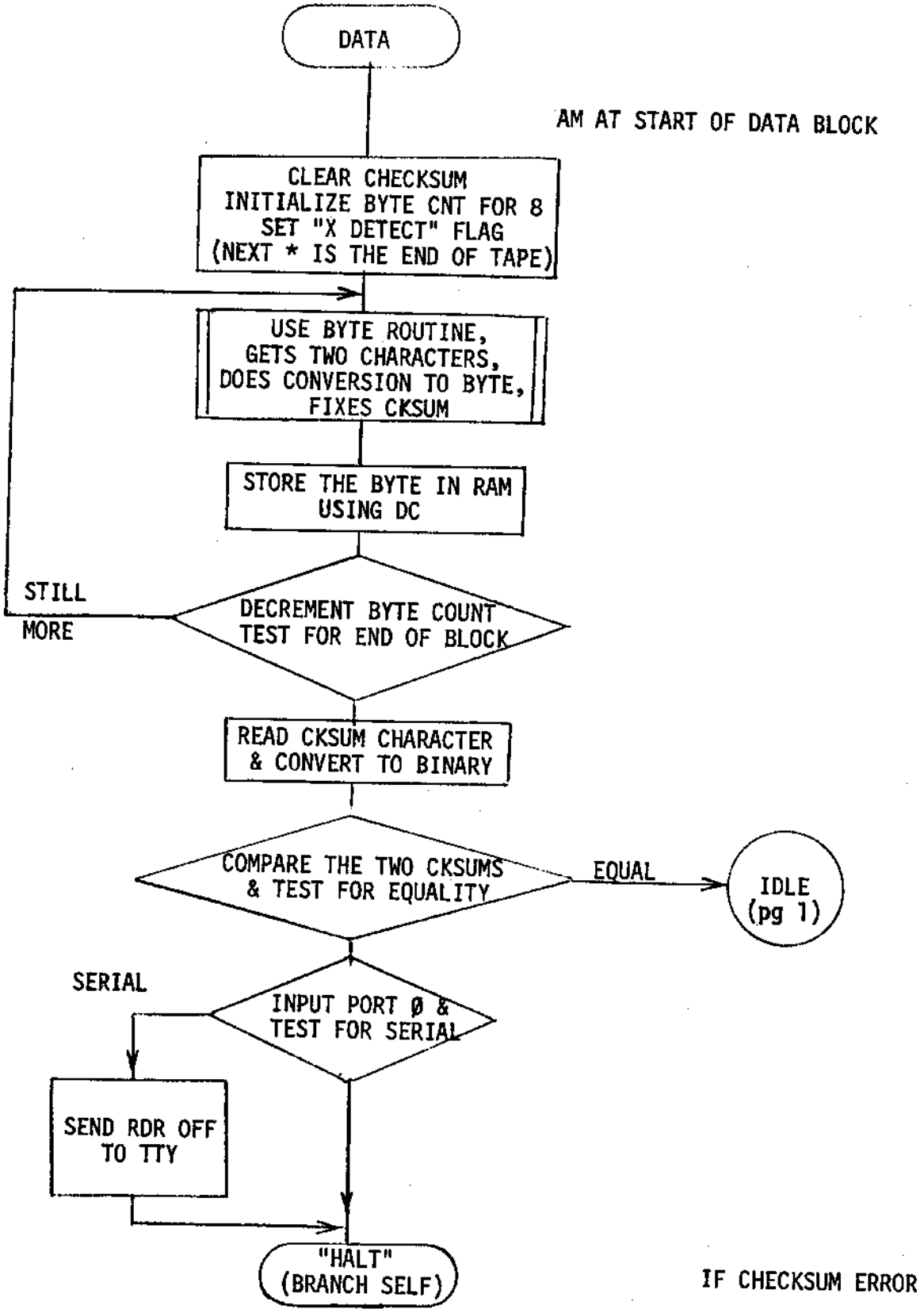
a 2μsec F8 system. The figure below shows the timing relationship of the step command and character ready status line.

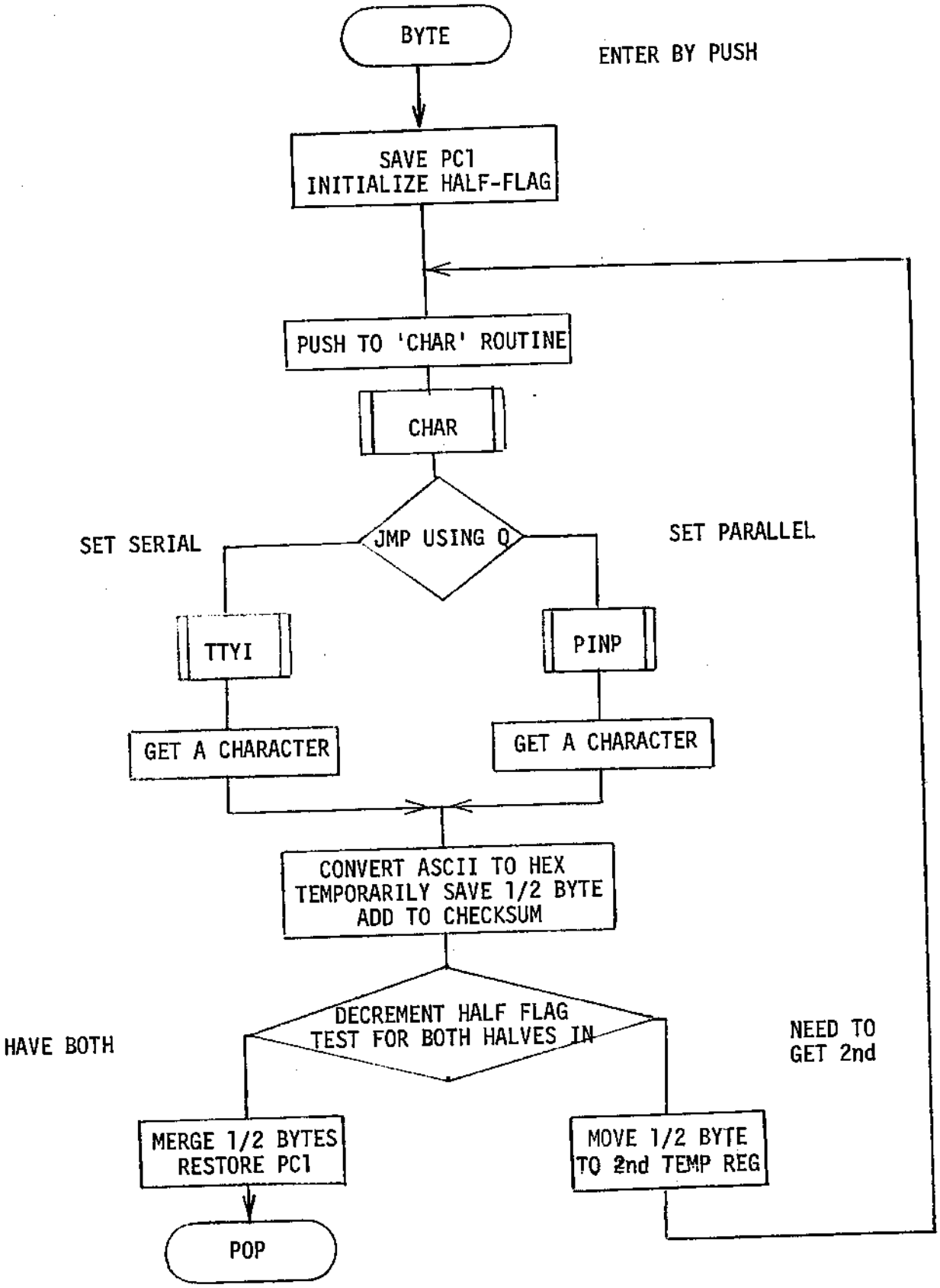


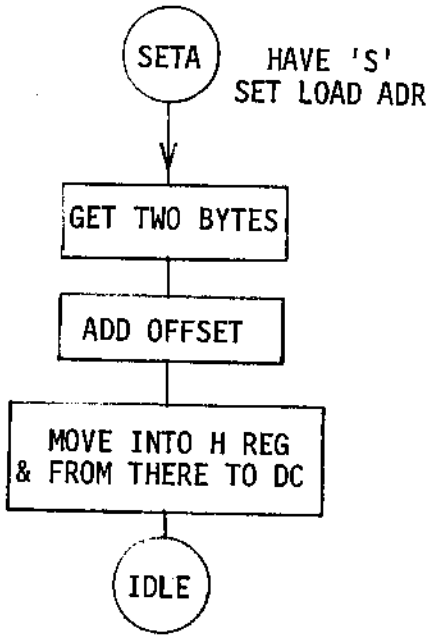
OPERATING INSTRUCTIONS:

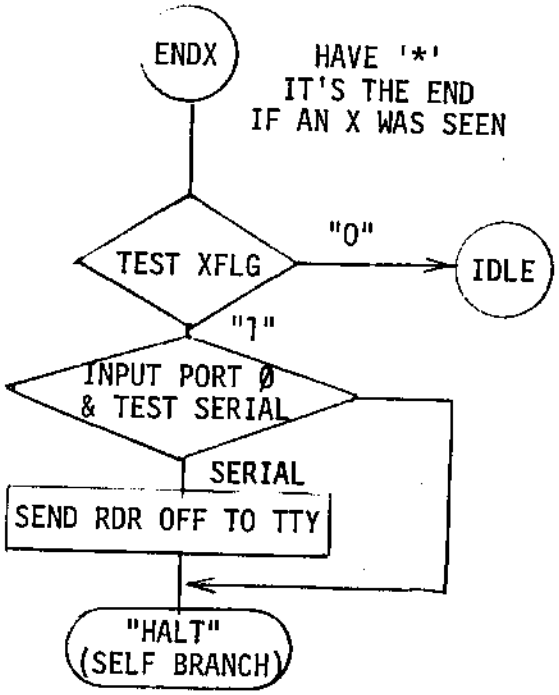
1. Disable the reader: for instance open the read head or set reader control to STOP or FREE.
2. Reset the system and start it running to initialize the I/O ports.
3. Load the tape into the reader anywhere before the start of the punching.
4. Again reset the system: the tape will begin moving as it is read into the system.
5. The system will "halt" at H'007C' when finished. A "halt" at H'0060' indicates a tape error; restart. A "halt" at H'0021' while using a parallel reader occurs if the reader is not ready. (The "halt" is a branch instruction branching to itself).

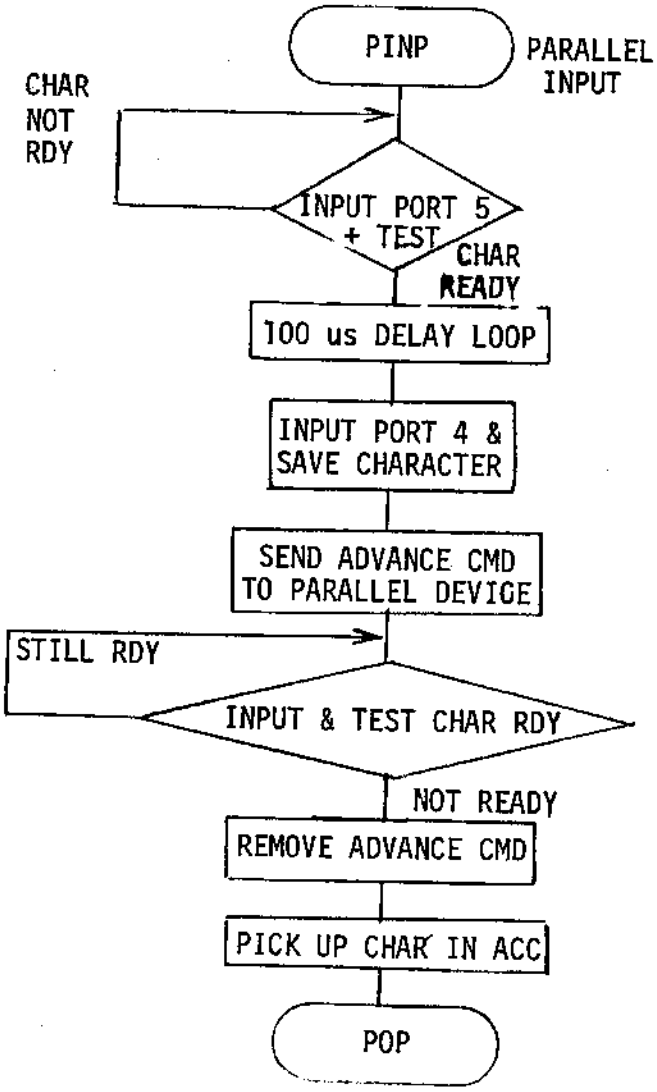

BOOT
INITIALIZE PORT Ø
INPUT & TEST
SERIAL
PARALLEL
SEND RDR "ON" TO TTY
SET SWITCH FOR PARALLEL INPUT (Q REG)
SET SWITCH FOR SERIAL INPUT
INITIALIZE PORT 4 & 5
INPUT & TEST FOR RDR READY
READY
NOT READY
"HALT" (BRANCH TO SELF)
SET DC TO DEFAULT, LOAD ADDR CLEAR "X DETECT" FLAG
IDLE
GET A CHARACTER FROM INPUT DEVICE
IS IT AN "S"?
SETA
IS IT A "*"?
ENDX
IS IT AN "X"?
DATA


DATA
AM AT START OF DATA BLOCK
CLEAR CHECKSUM
INITIALIZE BYTE CNT FOR 8
SET "X DETECT" FLAG
(NEXT * IS THE END OF TAPE)
USE BYTE ROUTINE,
GETS TWO CHARACTERS,
DOES CONVERSION TO BYTE,
FIXES CKSUM
STORE THE BYTE IN RAM
USING DC
STILL
MORE
DECREMENT BYTE COUNT
TEST FOR END OF BLOCK
READ CKSUM CHARACTER
& CONVERT TO BINARY
COMPARE THE TWO CKSUMS
& TEST FOR EQUALITY
EQUAL
IDLE
(pg 1)
SERIAL
INPUT PORT Ø &
TEST FOR SERIAL
SEND RDR OFF
TO TTY
"HALT"
(BRANCH SELF)
IF CHECKSUM ERROR


BYTE
ENTER BY PUSH
SAVE PC1
INITIALIZE HALF-FLAG
PUSH TO 'CHAR' ROUTINE
CHAR
JMP USING Q
SET SERIAL
SET PARALLEL
TTYI
PINP
GET A CHARACTER
GET A CHARACTER
CONVERT ASCII TO HEX
TEMPORARILY SAVE 1/2 BYTE
ADD TO CHECKSUM
DECREMENT HALF FLAG
TEST FOR BOTH HALVES IN
HAVE BOTH
NEED TO
GET 2nd
MERGE 1/2 BYTES
RESTORE PC1
MOVE 1/2 BYTE
TO 2nd TEMP REG
POP


SETA
HAVE 'S'
SET LOAD ADR
GET TWO BYTES
ADD OFFSET
MOVE INTO H REG
& FROM THERE TO DC
IDLE


ENDX
HAVE '*'
IT'S THE END
IF AN X WAS SEEN
TEST XFLG
"0"
IDLE
"1"
INPUT PORT Ø
& TEST SERIAL
SERIAL
SEND RDR OFF TO TTY
"HALT"
(SELF BRANCH)


PINP
PARALLEL
INPUT
CHAR
NOT
RDY
INPUT PORT 5
+ TEST
CHAR
READY
100 us DELAY LOOP
INPUT PORT 4 &
SAVE CHARACTER
SEND ADVANCE CMD
TO PARALLEL DEVICE
STILL RDY
INPUT & TEST CHAR RDY
NOT READY
REMOVE ADVANCE CMD
PICK UP CHAR IN ACC
POP

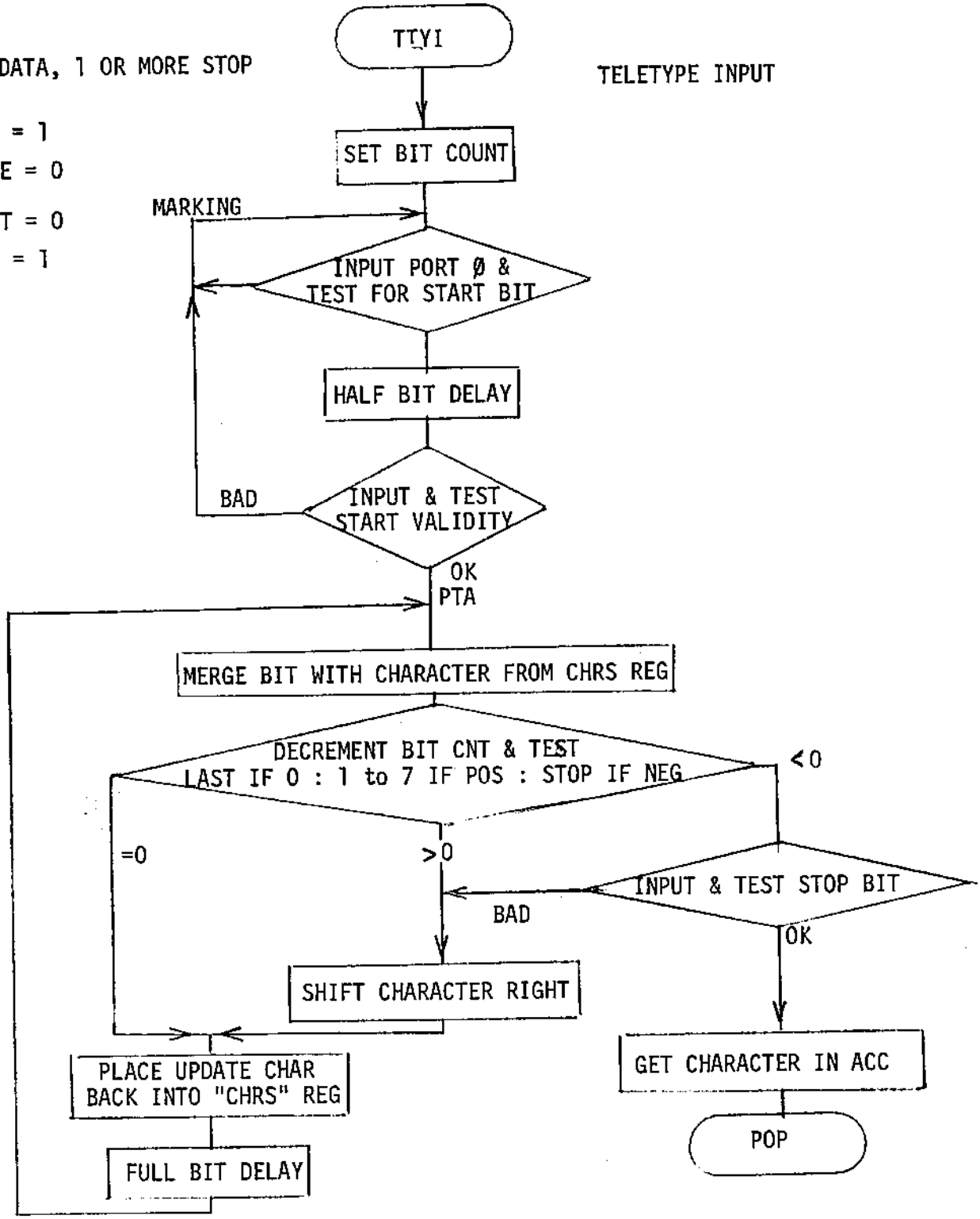
1 START, 8 DATA, 1 OR MORE STOP

MARK = 1

SPACE = 0

START = 0

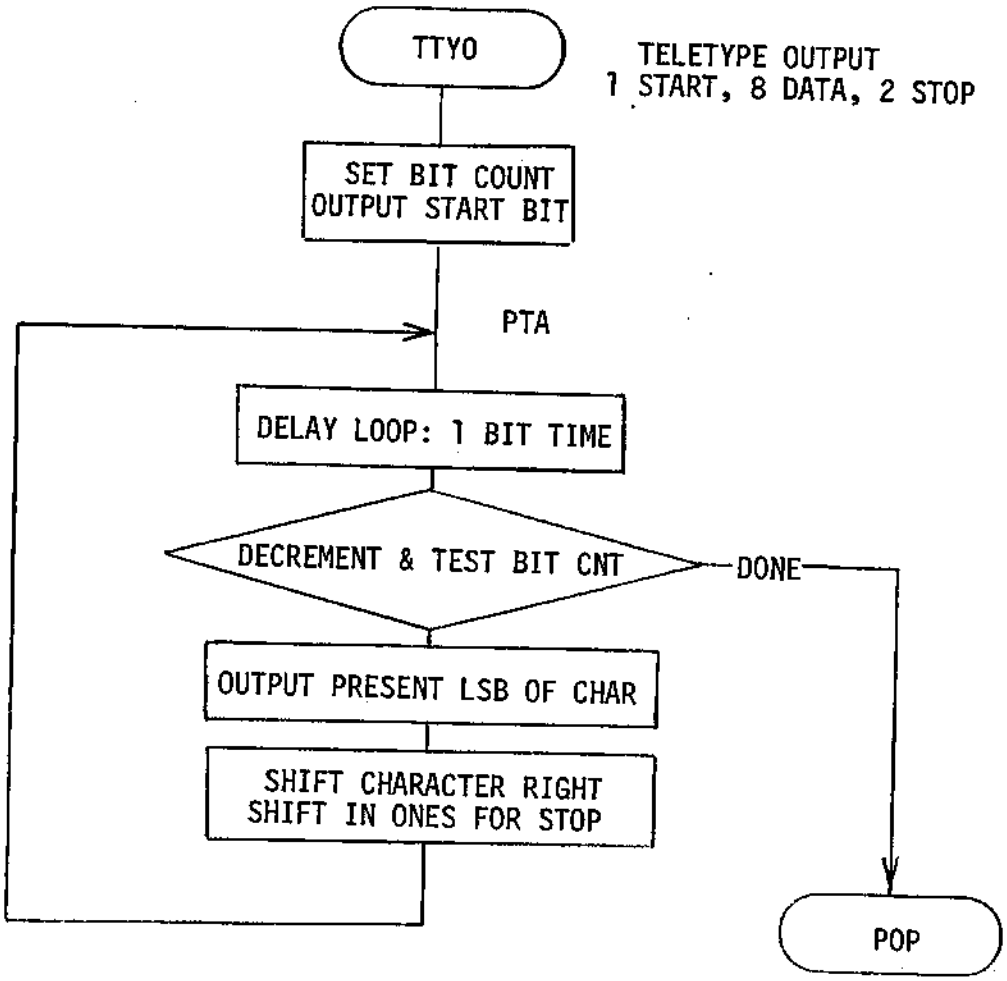
STOP = 1



SNAPSHOTS AT PT A

INPUT = "E" 0100 0101

BIT	BCNT	CONTENTS OF REG 'CHRS'
START	9	XXXX XXXX
D0—	8	0XXX XXXX
D1—	7	10XX XXXX
D2—	6	010X XXXX
D3—	5	1010 XXXX
D4—	4	0101 0XXX
D5—	3	0010 10XX
D6—	2	0001 010X
D7—	1	1000 1010
STOP	0	010[illegible]



CHARACTER & BIT COUNT AT PT A
IF INPUT IS "E" = 0100 0101

	CHAR	BIT CNT	OUTPUT
START	0100 0101	11	0
0	1010 0010	10	1
1	1101 0001	9	0
2	1110 1000	8	1
3	1111 0100	7	0
4	1111 1010	6	0
5	1111 1101	5	0
6	1111 1110	4	1
7	1111 1111	3	0
STOP 1	1111 1111	2	1
STOP 2	1111 1111	1	1

```
LOC   OBJECT CODE      STMT   SOURCE STATEMENT

PASS 1 COMPLETE    6.25 SEC
                         ******** BOOT PROGRAM FOR MMOD AND TTY OR READER AND FBA FORMAT****
                         *****          FOR EITHER SERIAL TTY OR PARALLEL TAPE READER ***
                         *****              SERIAL TTY WILL BE USED IF BIT 2 OF PORT 0 IS TIED LOW
                         *** FILE ALLOCATION          0: TEMP STORE FOR CHAR
                         *                            1: CHK SUM CHARACTER
                         *                            2: CHARACTER COUNT -- 16 CHARACTERS PER LIN
                         *                            3:  HALF BYTE FLAG
                         *                            4: TEMP STORE FOR 1ST HALF BYTE (UPPER)
                         *                            5:  X DETECT FLAG
                         *
                         ***** PORT ALLOCATION
                         *                            0: SERIAL TTY--  BIT 0: PNTR OUTPUT MARK=0V
                         *                                       BIT 3: TIE LOW TO CHOOSE TTY
                         *                                       BIT 7: KEYBRD INPUT MARK=0V
                         *                            4: PARALLEL BYTE INPUT
                         *                            5:  BIT 0--  STEP READER  0V=DRIVE LEFT
                         *                                BIT 6--  READER READY 5V=READY
                         *                                BIT 7--  SPOCKET  5V=CHAR READY
                         *
                         *
                         *
                1                BASE     HEX
                         *
                         *
                2                ORG      H'0000'
                3        BAUD    EQU      0
                4        CHRS    EQU      1
                5        BCNT    EQU      2
                6        TEMP    EQU      3
```

```
LOC  OBJECT CODE   STMT   SOURCE STATEMENT

                    7     XFLG    EQU     4
                    8     CONT    EQU     5
                    9     HFLG    EQU     6
                   10     CKSM    EQU     7
     0   71        11             LIS     01
     1   80        12             OUTS    0          INITIALIZE TTY PORT TO MARK STATE
     2   20   6    13             LI      H'06'
     4   50        14             LR      BAUD,A     SET DELAY TIME FOR TTY ROUTINES--
                        *                             3.3MS=300BPS, 9.09MS=110BPS
                        *
     5   A0        15             INS     0          INPUT SERIAL PORT
     6   15        16             SL      4
     7   81   E    17             BP      PBOO                  AND CHK FOR SERIAL TTY
                        *
     9   20   91   18             LI      H'91'      READER ON CHARACTER
     B   51        19             LR      CHRS,A     PASS IT
     C   26   0 DD 20             PI      TTYO
     F   20   B2   21             LI      TTYI                  AND TYPE IT
    11   7         22             LR      QL,A       SET Q REG FOR SERIAL INPUT CALLS
    12   70        23             LIS     0
    13   6         24             LR      QU,A
    14   90   E    25             BR      BOTI       SKIP OVER PARALLEL CODE
                        *
    16   20   9E   26     PBOO    LI      PINP
    18   7         27             LR      QL,A       SET Q FOR PARALLEL INPUT CALLLS
    19   70        28             LIS     0
    1A   6         29             LR      QU,A
    1B   B8        30             OUTS    8          INITIALIZE INPUT PORT
    1C   B9        31             OUTS    9          INITIALIZE CONTROL PORT FOR READER OFF
    1D   A9        32             INS     9
    1E   13        33             SL      1          GET TO READER READY BIT
    1F   81   3    34             BP      *+4
    21   90   FF   35     SLFI    BR      SLFI       LOCK UP IN LOOP IF READER NOT READY
                        *
                        *
    23   2A   8 0   36     BOTI    OCI     H'0800'    SET DC FOR LOADING ADDRESS
    26   70        37             LIS     0
    27   54        38             LR      XFLG,A     CLEAR FIRST X DETECT FLAG
                        *
                        *
    28   28   0 9D  39     IDLE    PI      CHAR       GET HEADER CHARACTER
    28   13        40             SL      1          CLEAR PARITY BIT
```

```
LOC   OBJECT CODE      STMT   SOURCE STATEMENT
2C    12               41          SR      1
2D    25  53           42          CI      C'S'    IS IT AN LOAD ADDRESS?
2F    84  32           43          BZ      SETA
31    25  2A           44          CI      C'*'    IS IT THE END OF THE TAPE?
33    84  36           45          BZ      ENDX
35    23  58           46          XI      C'X'    WELL, IF IT ISN'T AN X, THEN LET'S GO
37    94  F0           47          BNZ     IDLE
                       *
                       *
                       ******  HAVE THE START OF A DATA LINE  *******
                       *
39    57               48          LR      CKSM,A  INTIALIZE CHK SUM (ACC=0).
3A    78               49          LIS     H'08'
3B    55               50          LR      CCNT,A  INITIALIZE BYTE COUNT TO 8
3C    54               51          LR      XFLG,A  SHOW THAT X HAS BEEN DETECTED
                       *
3D    28  0  7E        52  CONT    PI      BYTE
40    17               53          ST              STORE THE BYTE
41    35               54          DS      CCNT    AND DECREMENT BYTE COUNT
42    94  FA           55          BNZ     CONT
                       *
                       ***                     NOW TIME FOR MAKING CKH SUM CHK
44    28  0  9D        56          PI      CHAR    GET CHK CHAR FROM TAPE
47    21  3F           57          NI      H'3F'   MASK TO SIX BITS
49    24  D0           58          AI      H'D0'   ASCII CONVERT-- FIRST 0-9
4B    82  3            59          BC      *+4     CARRY IF IT WAS THEEN A AND F
4D    24  39           60          AI      H'39'   FINISH CONVERSION OF A TO F
4F    18               61          COM
50    1F               62          INC
51    C7               63          AS      CKSM    MAKE THE COMPARISON
52    21  F            64          NI      H'0F'   MASK TO LSB 4 BITS
54    84  D3           65          BZ      IDLE    IF OK , LET'S GET SOME MORE
                       *
                       **********  CHK SUM ERROR HALT  ****************
56    A0               66          INS     0
57    15               67          SL      4       CHK FOR SER/PARALLEL
58    81  7            68          BP      SLF2    BIT IS NOT SET IF PARALLEL
5A    20  93           69          LI      H'93'   TURN READER OFF
5C    51               70          LR      CHRS,A
5D    28  0  DD        71          PI      TTYO            AND TYPE IT
60    90  FF           72  SLF2    BR      SLF2
```

```
LOC  OBJECT CODE     STMT   SOURCE STATEMENT
                       *
                       *
62   28  0  7E        73   SETA   PI    BYTE    GET NEW LOAD ADDRESS FROM TAPE
65   24  8            74          AI    H'08'   ADD IN OFFSET TO GET TO RAM
67   5A               75          LR    J0,A
68   28  0  7E        76          PI    BYTE
6B   5B               77          LR    I1,A
6C   10               78          LR    DC,H    SET THE ADDRESS INTO DC
6D   90  BA           79          BR    IDLE
                       *
                       *
6F   F4               80   ENDX   NS    XFLG    HAVE AN ?, BUT DOES IT FOLLOW ANY DATA ?
70   84  B7           81          BZ    IDLE    MUST BE ONLY AT THE HEAD OF THE TAPE
                       *                         WAS 2A+0  OR 2A+8
                       *********  HALT LOOP FOR WHEN FINISHED  ******
72   A0               82          INS   0
73   15               83          SL    4       MAKE TEST FOR SERIAL OR PARALLEL
74   81  7            84          BP    SLF3    SKIP OVER IF PARALLEL
76   20  93           85          LI    H'93'   READER OFF CHARACTER
78   51               86          LR    CHRS,A  PASS IT IN CHRS
79   28  0  DB        87          PI    TTYO
7C   90  FF           88   SLF3   BR    SLF3
                       *
                       *
                       *
                       *
                       *****  GET A BYTE  *********
                       **        GETS THE BYTE, CONVERTS, AND ADDS TO CHK SUM
                       *
7E   8                89   BYTE   LR    K,P     SAVE PC1
7F   72               90          LIS   2
80   56               91          LR    HFLG,A  SET THE HALF FLAG
81   28  0  9D        92   AGAIN  PI    CHAR
84   21  3F           93          NI    H'3F'   MASK TO 6 BITS
86   24  D0           94          AI    H'D0'   ASCII CONVERT-- FIRST 0-9
88   62  3            95          DC    **4
8A   24  39           96          AI    H'39'   NEXT CLEAN UP A-F
```

```
LOC   OBJECT CODE     STMT   SOURCE STATEMENT
8C    15              97          SL      4
8D    51              98          LR      CHRS,A     TEMP STORE IN REG CHRS
8E    14              99          SR      4
8F    C7              100         AS      CKSM       ADD NEW CHAR TO CHK SUM
90    57              101         LR      CKSM,A
91    36              102         DS      HFLG       DECREMENT HALF COUNT
92    84   5          103         BZ      TWOD       HAVE BOTH HALVES
                            *
94    41              104         LR      A,CHRS     ONLY HAVE UPPER HALF-- MUST GO BACK
95    53              105         LR      TEMP,A           BUT FIRST SAVE THIS HALF AWAY
96    90   EA         106         BR      AGAN
                            *
98    41              107   TWOD  LR      A,CHRS     GET THE LAST CHAR READ
99    14              108         SR      4                AND SHIFT IT TO LSB END
9A    C3              109         AS      TEMP       ADD IN MSB HALF BYTE
9B    9               110         LR      P,K        RESTORE PCL
9C    1C              111         POP
                            *
```

```
LOC  OBJECT CODE     STMT  SOURCE STATEMENT
                           *
                           *
                           ********     PARALLEL AND SERIAL INPUT ROUTINES     ********
                           **
                           **      COMMON CALL IS A PUSH TO CHAR
                           **      CHAR USES Q TO JUMP TO APPROPRIATE ROUTINE
                           **
                           **
 9D   D                 113  CHAR   LR      PQ,Q     JUMP TO INPUT ROUTINE INDIRECTLY THRU Q REG
                           *
                           *
                           *
                           **  PINP:  GET A CHARACTER FROM PORT .
                           **
                           **   TYPICALLY USED WITH TAPE READER, BUT HAS A HANDSHAKING
                           **   DISCIPLINE THAT IS APPLICABLE TO OTHER DEVICES SUCH AS FIFOS
                           **   . LOOKS FOR A CHARACTER READY INPUT, AND THEN GETS THE
                           **   CHARACTER. NEXT CPU GIVES AN ADVANCE PULSE THAT IS REMOVED
                           **   AFTER THE DEVICE READY INPUT GOES NOT READY.
                           **
                           **
 9E   A9                114  PINP   INS     9        GET A CHAR FROM 300CPS READER
 9F   91  FE            115         BM      PINP     LOOK FOR SPOCKET= HIGH
 A1   7F                116         CIS     H'0F'    100 US DELAY AFTER SEEING SPOCKET
 A2   18                117         COM
 A3   1F                118  POLY   INC
 A4   94  FE            119         BNZ     POLY
                           *
 A6   A8                120         INS     8                AND NOW GET THE DATA BYTE
```

```
LOC  OBJECT CODE   STMT   SOURCE STATEMENT

A7   16            121          COM
A8   51            122          LR      CHRS,A   TEMP STORE CNEW CHAR
                        *
A9   71            123          LIS     H'03'    LET'S ADVANCE THE READER TO NEXT CHAR
AA   B9            124          OUTS    9
AB   A9            125  NOSP    INS     9        GET READER STATUS
AC   81  FE        126          BP      NOSP           AND LOOK FOR MOVING OFF SPOCKET
AE   70            127          LIS     0
AF   B9            128          OUTS    9        REMOVE DRIVE PULSE NOW THAT IT IS MOVING
                        *
B0   41            129          LR      A,CHRS   PICK BACK UP THE NEW CHARACTER
B1   1C            130          POP
                        *
                        *
                        *
                        *** TTYI:  SERIAL INPUT        CHARACTER RETURNED IN ACC AND REG0
                        *
                        **   REG BCNT HOLDS BIT COUNT . REG CHRS HOLDS CHARACTER
B2   79            131  TTYI    LIS     9
B3   52            132          LR      BCNT,A   SET BIT COUNT FOR 8 DATA BITS
B4   A0            133  START   INS     0
B5   91  FE        134          BM      START    LOOK FOR START BIT
B7   40            135          LR      A,BAUD   GET DELAY COUNT
B8   28            136          NOP
B9   90  1         137  DLY3    BR      *+2      SILLY BRANCH FOR DELAY
BB   24  1         138          AI      H'01'
BD   94  FB        139          BNZ     DLY3     THIS LOOP IS HALF AS MUCH DELAY
                        *
BF   A0            140          INS     0        CHECK START BIT VALIDITY
C0   91  F3        141          BM      START
C2   21  80        142  LOOP    NI      H'80'    MASK TO GET INPUT BIT ONLY
C4   C1            143          AS      CHRS     (LD'G START BIT WILL CLR GARBAGE)
C5   32            144          DS      BCNT     DROP BIT CNT: 0 IF LAST DATA,NEG IF STOP
C6   91  11        145          BM      STOP     NEG IF LOOKING FOR STOP BIT
C8   84  2         146          BZ      LOP2     IF LAST BIT, DO NOT SHIFT
CA   12            147  LOP3    SR      1        SHIFT ASSEMBLED CHARACTER TO MAKE ROOM
CB   51            148  LOP2    LR      CHRS,A   STORE ASSEMBLED CHARACTER
                        *
CC   40            149          LR      A,BAUD   START OF FULL BIT TIME DELAY
CD   28            150          NOP
```

```
LOC  OBJECT CODE     STMT  SOURCE STATEMENT
CE   8B              151   DLY4   OUTS   H'0B'   NOP FOR DELAY
CF   8B              152          OUTS   H'0B'
D0   8B              153          OUTS   H'0B'
D1   24   1          154          AI     H'01'   INCR WITH A 5US INST
D3   94   FA         155          BNZ    DLY4

                          *
D5   A0              156          INS    0       GET NEW BIT
D6   90   EB         157          BR     LOOP

                          *
D8   A0              158   STOP   INS    0       GET STOP BIT
D9   41              159          LR     A,CHRS  GET CHAR IN ACC
DA   81   EF         160          BP     LOP3    BAD STOP BIT IF BIT7 OF INPUT IS 0

                          *
DC   1C              161          POP

                          *
```

```
LOC  OBJECT CODE      STMT   SOURCE STATEMENT

                       **
                       **
                       ******* SERIAL OUTPUT ROUTINE  ***********
                       **
                       **  HAS 3 START, 8 DATA, 2 STOP. USES PORT 0, BITS 0 AND 7
                       **  BAUD RATE IS SET BY A DELAY COUNT IN REG BAUD
                       **   CALL BY PUTTING CHAR IN REG CHRS, AND SETTING DELAY IN BAUD
                       **   ROUTINE RETURNS WITH ALL 1'S IN REG CHRS, REG BAUD INTACT
                       **
                       **
 DD   7B               163   TTY0   LIS    H'0B'
 DE   52               164          LR     BCNT,A    SET BIT COUNT FOR 11 BITS
 DF   70               165          LIS    0
 E0   B0               166          OUTS   0         OUTPUT START BIT
                       *
                       ***   DELAY ROUTINE-- 3.3MS FOR 300 BAUD, 9 MS FOR 110 BAUD
                       *
 E1   40               167   DLY1   LR     A,BAUD    GET DELAY COUNT
 E2   2B               168          NOP
 E3   BB               169   DLY2   OUTS   H'0B'     NOP FOR DELAY
 E4   BB               170          OUTS   H'0B'
 E5   BB               171          OUTS   H'0B'
 E6   24   1           172          AI     H'01'     INCR WITH A 5US INST
 E8   94   FA          173          BNZ    DLY2
                       *
 EA   32               174          DS     BCNT      DECREMENT BIT COUNT
 EB   84   C           175          BZ     DONE
 ED   41               176          LR     A,CHRS    GET CHARACTER
 EE   21   1           177          NI     H'01'     MASK OFF ALL BUT BIT 0
 F0   B0               178          OUTS   0         OUTPUT THE NEW DATA BIT
                       *
 F1   41               179          LR     A,CHRS    NOW SHIFT THE CHARACTER FOR NEXT BIT
 F2   12               180          SR     1
 F3   24   80          181          AI     H'80'     FILL WITH 1'S FOR STOP BITS
 F5   51               182          LR     CHRS,A
```

```
LOC   OBJECT CODE    STMT   SOURCE STATEMENT
F6    90 EA          183          BR      DLY1     GO WAIT OUT THIS BIT
                            *
F8    1C             184    DONE  POP                ALL FINISHED
                            *
                            *
                            *
                     185          END
PASS 2 COMPLETE    0.17 SEC
NO ERRORS IN ABOVE ASSEMBLY
```